



Microsoft Azure Fundamentals

AZ-900

Azure Fundamentals

300 Interview Questions & Answers

Complete preparation guide for AZ-900 certification exam

8 Sections | 300 Q&A | User-Friendly Explanations in Hinglish

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Sec 1: Cloud Concepts

Sec 2: Compute

Sec 3: Storage

Sec 4: Networking

Sec 5: Identity & Security

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Sec 8: Tools & More

8 comprehensive sections covering all AZ-900 exam topics

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About this Guide

This comprehensive study guide covers all major topics of the Microsoft Azure Fundamentals AZ-900 certification exam. Questions are written in simple, user-friendly language so every concept is clear and easy to remember. Each answer explains the concept practically with real-world examples.

How to Use: Read each question first, attempt your answer, then check the explanation. Mark difficult questions and revise them before the exam. Good luck!

Section 1: Cloud Concepts

Q1. Cloud Computing kya hota hai? Simple language mein samjhao.

Cloud computing matlab hai internet ke through computers, servers, storage aur software ka use karna – apne paas hardware rakhe bina. Jaise Netflix streaming, aap apne device par install nahi karte, internet se seedha dekhte ho. Waise hi cloud mein resources 'rent' par milte hain, on-demand basis par.

Q2. Cloud ke 3 main deployment models kaunse hain?

1. Public Cloud – Microsoft Azure, AWS jaise providers ke servers sab ke liye available hote hain (shared infrastructure).
2. Private Cloud – Sirf ek organization ke liye dedicated resources (on-premises ya hosted).
3. Hybrid Cloud – Public + Private dono ka combination. Sensitive data private mein, baaki public mein.

Q3. Azure ka full form kya hai aur ye Microsoft ka kya product hai?

Azure ka koi official full form nahi hai – yeh ek brand name hai. Azure wo color hota hai jo clear blue sky ka hota hai. Microsoft Azure ek public cloud computing platform hai jo 200+ services provide karta hai jaise Virtual Machines, Databases, AI, Networking, Storage, etc.

Q4. CapEx aur OpEx mein kya difference hai?

CapEx (Capital Expenditure) – Pehle se bada investment karo, jaise server kharidna. Yeh ek baar ka cost hai.

OpEx (Operational Expenditure) – Jitna use karo utna pay karo (pay-as-you-go). Cloud mein mostly OpEx model follow hota hai, jisse companies flexible rahti hain aur budget plan karna easy ho jaata hai.

Q5. Consumption-based model kya hota hai?

Is model mein aap sirf wahi resources pay karte ho jo aapne actually use kiye. Jaise electricity ka bill – aapne jitni units jalayi, utna hi pay kiya. Azure mein bhi aap VM chalte ho toh hi pay karte ho, band karo toh cost zero. Isse waste kam hota hai aur efficiency badhti hai.

Q6. Scalability aur Elasticity mein kya difference hai?

Scalability – System ka capacity badhane ki ability (manually ya planned). Jaise ek website ke liye zyada servers add karna.

Elasticity – System automatically demand ke hisaab se resources badhata-ghataata hai. Jaise festival sale pe automatically zyada servers on ho jaana aur baad mein automatically band ho jaana.

Q7. High Availability ka matlab kya hai?

High Availability (HA) matlab hai system 99.9%+ time available/running rahe — downtime minimum ho. Azure mein SLA (Service Level Agreement) ke through guarantee milti hai. Multiple data centers, redundant systems aur auto-failover se HA achieve hoti hai. Example: Azure VM ka SLA 99.99% hai agar 2+ VMs different Availability Zones mein hon.

Q8. Fault Tolerance aur Disaster Recovery mein kya antar hai?

Fault Tolerance – System failure hone ke BAWAJOOD kaam karta rahe. Jaise airplane mein backup engine.

Disaster Recovery – Agar sab kuch fail ho jaye (flood, earthquake), toh data/system kaise recover karein. Azure mein Azure Site Recovery (ASR) iska tool hai. Fault tolerance real-time hai, DR ek plan/process hai.

Q9. IaaS, PaaS aur SaaS kya hote hain — example ke saath samjhao.

IaaS (Infrastructure as a Service) – Sirf hardware milta hai virtual form mein (VM, Storage, Network). Aap OS, apps sab install karo. Example: Azure Virtual Machines.

PaaS (Platform as a Service) – Platform milta hai deploy karne ko, infrastructure ki chinta nahi. Example: Azure App Service.

SaaS (Software as a Service) – Ready-made software milta hai. Example: Microsoft 365, Gmail.

Q10. Shared Responsibility Model kya hota hai Azure mein?

Is model mein kuch cheezein Microsoft ki responsibility hain, kuch aapki.

Microsoft ki: Physical security, hardware, network infrastructure, hypervisor.

Aapki: Data, identity management, user access, application security.

Middle mein (depends on service type): OS updates, network config. PaaS mein Microsoft zyada handle karta hai, IaaS mein aap zyada.

Q11. Azure Regions kya hain? Kitne hain approximately?

Azure Region ek geographic area hai jisme Microsoft ke data centers hain. Ye regions globally distributed hain — 60+ regions in 140+ countries. Example: East US, West Europe, Central India, Southeast Asia. Aap choose karte ho ki aapka data/service kahan host ho — performance, compliance aur pricing ke basis par.

Q12. Availability Zones kya hoti hain?

Availability Zones ek region ke andar physically separate data centers hain — alag power supply, cooling, networking ke saath. Agar ek zone fail ho, doosra kaam karta rahe. Minimum 3 zones per region hoti hain. Mission-critical applications ke liye use hota hai. Azure VM agar 2+ zones mein ho toh 99.99% SLA milti hai.

Q13. Azure Region Pairs kya hain?

Azure mein har region ka ek 'paired region' hota hai — generally 300+ miles door. Agar koi disaster ek region mein aaye, paired region se recovery ho sake. Example: East US aur West US paired hain. Microsoft updates bhi ek baar mein sirf ek pair ka ek region update karta hai, taki downtime na aaye.

Q14. Azure Geography kya hoti hai?

Geography ek specific market/area hai jo data residency aur compliance boundaries define karta hai. Example: 'India' geography mein Central India, South India, West India regions hain. Isse companies ko apna data specific country mein rakhne ki guarantee milti hai — jo legal ya government requirements ke liye important hai.

Q15. Benefits of Cloud Computing kya hain?

1. Cost Savings – No upfront hardware cost.
2. Scalability – Zaroorat ke hisaab se badhao-ghataao.
3. Reliability – High availability, backup, disaster recovery.
4. Security – Enterprise-grade security Microsoft handle karta hai.
5. Speed – Minutes mein resources ready.
6. Global Reach – Duniya mein kahin bhi deploy karo.
7. Innovation – AI, ML, IoT latest tech instantly available.

Q16. Cloud mein Agility ka kya matlab hai?

Agility matlab hai quickly respond karna business changes ke liye. Cloud mein naye servers, databases ya applications minutes mein launch ho jaate hain — pehle hardware order karne mein weeks lagte the. Isse startups se lekar enterprises tak sab apne products fast deploy aur test kar sakte hain. Time-to-market dramatically reduce ho jaata hai.

Q17. Azure free account mein kya milta hai?

Azure Free Account mein milta hai:

- \$200 credit pehle 30 din ke liye.
- 12 mahine ke liye select services free (e.g., 750 hours B1S VM, 5 GB Blob Storage).
- Hamesha ke liye free services jaise Azure Functions (1 million calls/month), Cosmos DB (1000 RU/s), App Service (10 web apps).

Ye students aur beginners ke liye practice ke liye bahut useful hai.

Q18. Azure Portal kya hai?

Azure Portal (portal.azure.com) ek web-based graphical interface hai jahan se aap apne sare Azure resources manage kar sakte ho — bina command line jaane. Drag-and-drop style mein VMs create karo, databases configure karo, billing check karo, alerts set karo. Mobile app bhi available hai. Beginners ke liye sabse easy starting point hai.

Q19. Azure CLI aur Azure PowerShell mein kya farak hai?

Azure CLI – Command-line tool jo Linux, Mac, Windows sabpar chalta hai. Commands 'az' se start hoti hain. Example: az vm create.

Azure PowerShell – PowerShell cmdlets use karta hai, Windows users ke liye familiar. Example: New-AzVM.

Dono se same kaam hota hai – automation aur scripting ke liye. CLI cross-platform users prefer karte hain, PowerShell Windows admins.

Q20. Azure Resource Manager (ARM) kya hai?

ARM Azure ka deployment aur management layer hai. Jab bhi aap Azure Portal, CLI ya PowerShell se kuch create karte ho, request ARM ke through jaati hai. ARM ensure karta hai consistent management, access control (RBAC), tagging, aur resource grouping. ARM Templates (JSON/Bicep files) se infrastructure-as-code possible hai.

Q21. Resource Group kya hota hai?

Resource Group ek logical container hai jisme related Azure resources rakhe jaate hain – jaise ek project ke saare resources ek group mein. Ek saath manage, delete, monitor karne ke liye useful. Example: ek web app ke liye VM + Database + Storage ek resource group mein. Delete resource group = sab kuch ek baar mein delete.

Q22. Azure Subscription kya hoti hai?

Subscription ek billing unit hai Azure mein – isme resources create hote hain aur cost track hoti hai. Ek Azure account mein multiple subscriptions ho sakti hain. Example: Dev subscription alag, Production subscription alag – taki costs aur access separate rahein. Enterprise mein department-wise subscriptions hoti hain.

Q23. Management Groups kya hain?

Management Groups subscriptions ko organize karne ka tarika hai. Agar company mein 50 subscriptions hain, toh unhe logically group kar sako (e.g., HR Department, IT Department). Policies aur RBAC parent management group par apply karo – automatically sari child subscriptions pe lagu ho jata hai. Hierarchy: Management Groups > Subscriptions > Resource Groups > Resources.

Q24. Azure cost management ke liye kaunse tools hain?

1. Azure Cost Management + Billing – Usage aur spending track karo.
 2. Budgets – Alert set karo jab spending limit cross ho.
 3. Azure Advisor – Cost optimization recommendations.
 4. Azure Pricing Calculator – Deploy karne se pehle cost estimate karo.
 5. TCO Calculator – On-premises vs Azure cost compare karo.
- Ye sab tools free available hain Azure portal mein.

Q25. TCO Calculator kya hota hai?

TCO = Total Cost of Ownership. Azure TCO Calculator se aap compare kar sakte ho ki apna on-premises infrastructure maintain karna kitna expensive hai vs Azure par shift karna. Aap apni current servers, storage, networking details fill karte ho aur tool 3-5 year ka comparison dikhata hai. Mostly Azure mein significant savings milti hain.

Q26. SLA (Service Level Agreement) kya hota hai?

SLA ek formal commitment hai Microsoft ka — ki Azure service kitna % uptime guarantee karta hai. Example: Azure VM (single instance) = 99.9%, Azure VM (2+ instances in availability set) = 99.95%, Azure VM (2+ Availability Zones) = 99.99%. Agar SLA miss ho, customers ko service credits milte hain compensation ke roop mein.

Q27. Composite SLA kya hota hai?

Jab multiple services ek saath use hoti hain, toh composite SLA calculate hoti hai — multiply karke. Example: Web App SLA 99.95% + Database SLA 99.99% = $0.9995 \times 0.9999 = 99.94\%$. Isliye distributed systems design karte waqt SLA calculations important hain. Redundancy add karne se composite SLA better hoti hai.

Q28. Azure ka pricing model samjhao.

Azure mein multiple pricing tiers hain:

1. Pay-as-you-go – Jitna use utna pay.
2. Reserved Instances – 1-3 saal ka commitment, 40-70% savings.
3. Spot Instances – Unused capacity, bahut sasta lekin interrupt ho sakta hai.
4. Azure Hybrid Benefit – Existing Windows/SQL licenses use karke aur savings.
5. Free Tier – Limited free services hamesha.

Best savings = Reserved + Hybrid Benefit combine karo.

Q29. What is the difference between Vertical and Horizontal Scaling?

Vertical Scaling (Scale Up/Down) – Ek machine ki power badhao. Jaise 4 CPU se 8 CPU karo, ya RAM 16GB se 32GB karo. Ek limit hoti hai is approach ki.

Horizontal Scaling (Scale Out/In) – Zyada machines add karo. Jaise 2 VMs ki jagah 10 VMs chadha do load badhne par. Cloud mein horizontal scaling preferred hai kyunki practically unlimited hai.

Q30. Predictive vs Reactive Scaling kya hota hai?

Reactive Scaling – Current load dekh kar scale karo (jab CPU 80% ho jaaye tab scale out).

Predictive Scaling – Historical patterns se AI predict karta hai future load aur advance mein scale karta hai. Example: Har Monday morning traffic spike hoti hai, toh Sunday raat ko hi scale out kar do. Azure Virtual Machine Scale Sets dono support karte hain.

Q31. Azure mein Serverless computing kya hai?

Serverless matlab servers toh hain background mein, lekin aapko manage nahi karne. Aap sirf code likhte ho, Azure execute karta hai on-demand. Example: Azure Functions — ek function trigger hota hai event se, runs, aur band ho jaata hai. Pay-per-execution model hai. Servers ka provision/scaling sab automatic. Small tasks ke liye extremely cost-effective.

Q32. Edge Computing kya hai Azure context mein?

Edge Computing matlab processing user/device ke paas karna — cloud data center tak jaane ki bajaye. Azure mein: Azure IoT Edge devices par processing hoti hai. Isse latency kam hoti hai — jaise factory machines mein real-time decisions lene ke liye data cloud bhejna wait karna expensive ho sakta hai. Edge par process karke locally decide karo.

Q33. What is Cloud Bursting?

Cloud Bursting ek hybrid cloud strategy hai. Normal time par private cloud use karo. Jab peak load aaye (jaise Diwali sale), automatically public cloud (Azure) mein overflow karo. Isse private cloud overprovisioned nahi karna padta aur peak demands bhi handle ho jaati hain. Best of both worlds.

Q34. Azure mein Multicloud kya hota hai?

Multicloud matlab ek se zyada cloud providers use karna — Azure + AWS + Google Cloud. Reasons: vendor lock-in avoid karna, best services choose karna (e.g., GCP ka AI best, Azure ka hybrid best). Azure Arc tool hai jo Azure se multiple clouds ko manage karne deta hai ek single pane of glass se.

Q35. What are the characteristics of Cloud Computing per NIST?

NIST ne 5 essential characteristics define ki hain:

1. On-demand self-service – User khud resources provision kare.
2. Broad network access – Internet se anywhere accessible.
3. Resource pooling – Multiple users share resources.
4. Rapid elasticity – Quick scale up/down.
5. Measured service – Usage track aur bill hota hai.

Az-900 exam mein ye important hain.

Q36. Azure mein Compliance kyu important hai?

Companies ko government aur industry regulations follow karni hoti hain — jaise GDPR (Europe data privacy), HIPAA (US healthcare), ISO 27001. Azure 90+ compliance certifications maintain karta hai. Azure Trust Center aur Compliance Manager se aap dekh sakte ho ki aapki services compliant hain ya nahi. Government/finance/healthcare industries ke liye critical hai.

Q37. What is Azure Sovereign Cloud?

Sovereign Clouds specific governments ke liye isolated Azure environments hain — regular Azure se completely separate.

- Azure Government (US government)
- Azure China (Operated by 21Vianet, Chinese regulations ke liye)
- Azure Germany (German data sovereignty ke liye, ab mostly integrated)

In mein data us specific country mein rehta hai, aur sirf authorized users access kar sakte hain.

Q38. What is meant by 'Economies of Scale' in cloud?

Economies of Scale matlab hai bade scale par kaam karne se per-unit cost kam ho jaata hai. Microsoft globally lakho servers chalata hai — bulk mein hardware kharidta hai, energy efficient data centers banata hai. Iska faida customers ko milta hai: aap apne resources ke liye kam pay karte ho compared to khud same infrastructure banana. Cloud providers is advantage ko customers tak pass-through karte hain.

Q39. What is Azure Arc?

Azure Arc ek service hai jo aapko Azure management capabilities extend karne deta hai — on-premises servers, Kubernetes clusters, aur even other cloud providers (AWS, GCP) par. Matlab: Azure portal se apne non-Azure resources bhi manage kar sako, same policies apply karo, same security use karo. Hybrid aur multicloud management ke liye key tool hai.

Q40. Microsoft Azure ka history brief mein batao.

Azure 2008 mein 'Project Red Dog' ke naam se shuru hua, 2010 mein publicly launch hua 'Windows Azure' naam se. 2014 mein 'Microsoft Azure' rename hua. Aaj 60+ regions, 200+ services hain. Cloud market mein AWS ke baad second largest provider hai, aur enterprise market mein AWS se competitive hai. India mein Central India, South India, West India regions hain.

Q41. What is the Azure global infrastructure?

Azure global infrastructure mein shamil hain:

- 60+ Regions worldwide
- 3+ Availability Zones per region (where available)
- 116+ Edge locations (Azure CDN)
- Dedicated undersea fiber cables globally
- 200+ physical data centers

India mein 3 regions hain: Central India (Pune), South India (Chennai), West India (Mumbai).

Q42. Describe the concept of 'Trust but Verify' in Cloud Security.

Modern approach hai 'Zero Trust' — kabhi trust mat karo, hamesha verify karo. Matlab sirf VPN ke andar hona security guarantee nahi karta. Har request ko authenticate karo, least privilege access do, network location par blindly trust mat karo. Azure mein Conditional Access, MFA, Identity Protection ye sab Zero Trust implement karne ke tools hain.

Q43. What is latency and how does cloud reduce it?

Latency = data travel karne ka time (milliseconds mein). Jaise Mumbai se US server tak data bhejne mein ~200ms lag sakta hai. Cloud providers ne duniya bhar mein data centers banaye hain — aap apni service nearest region mein deploy karte ho, users ko kam latency milti hai. Azure CDN aur Azure Front Door aur bhi optimize karte hain.

Q44. Cloud mein 'Pay-as-you-go' model ke kya benefits hain?

1. No upfront investment – Pehle paise kharch nahi karne.
 2. Match spending to actual usage – Kam use karo toh kam pay karo.
 3. Experiment freely – New services try karo, fail karo, seekho — cost minimal.
 4. Predictable billing – Usage track karke budget plan karo.
 5. No over-provisioning – 10 servers kharid ke 2 nahi chalane.
- Startups aur projects ke liye game-changer hai ye model.

Q45. What is provisioning in cloud context?

Provisioning = resources ready karna use ke liye. Traditional mein: server order karo, wait karo, install karo — weeks lagte the. Cloud mein: portal ya CLI se click karo, minutes mein server ready. Self-service provisioning cloud ki core feature hai. Over-provisioning = zyada resources lena (waste), Under-provisioning = kam resources lena (performance issue).

Q46. What is a cloud service model comparison?

IaaS – Aap manage karte ho: OS, Runtime, Apps, Data. Azure manages: Virtualization, Servers, Storage, Networking.
PaaS – Aap manage karte ho: Apps, Data. Azure manages: Everything else including OS, runtime.
SaaS – Aap manage karte ho: Just your data and user access. Azure/Vendor manages: Everything.
Jitna upar jaao, utni zyada responsibility Azure ki, utni zyada aapki freedom.

Q47. What is serverless pricing model?

Serverless (Azure Functions) mein pricing hoti hai per execution basis par:
– First 1 million executions free per month
– Uske baad ~\$0.20 per million executions
– Memory + Duration ke basis par bhi calculate hoti hai
Iska faida: agar function sirf 1000 baar chale mahine mein, toh practically free. Traditional server chalaane ka fixed cost nahi lagta.

Q48. Cloud adoption challenges kya hain?

1. Security concerns – Data cloud par rakhna safe hai kya?
 2. Compliance – Regulations follow ho rahe hain kya?
 3. Cost management – Unexpected bills aa sakte hain.
 4. Skill gap – Team ko cloud skills nahi hoti.
 5. Vendor lock-in – Ek provider se doosre par shift karna mushkil.
 6. Latency – Kuch applications ko on-premises better performance milti hai.
- Azure in sab ke solutions provide karta hai training, tools aur services ke through.

Q49. What are the types of cloud users?

1. Individual users – Personal projects, learning.
 2. Startups – Fast scaling, minimal upfront cost.
 3. SMBs (Small/Medium Businesses) – Affordable enterprise-grade services.
 4. Enterprises – Large-scale deployments, compliance, hybrid.
 5. Government – Sovereign clouds, compliance requirements.
 6. Developers – DevOps, CI/CD, testing environments.
- Azure sab ke liye solutions banata hai – free tier se enterprise agreements tak.

Q50. Summarize the key points of AZ-900 Cloud Concepts.

- AZ-900 Cloud Concepts mein ye main topics cover hote hain:
1. Cloud benefits: cost savings, scalability, reliability, security, global reach.
 2. Cloud service types: IaaS, PaaS, SaaS.
 3. Cloud deployment: Public, Private, Hybrid.
 4. Consumption model: CapEx vs OpEx, pay-as-you-go.
 5. Shared responsibility model.
 6. Azure global infrastructure: regions, availability zones, pairs.
- Ye sab exam mein ~20-25% weightage rakhte hain.

Section 2: Azure Core Services – Compute

Q51. Azure Virtual Machines kya hoti hain?

Azure Virtual Machine (VM) ek cloud mein virtual computer hai — jaise aapka laptop/desktop lekin Microsoft ke data center mein. Aap choose karte ho OS (Windows/Linux), CPU, RAM, Storage. Remote access hoti hai (RDP ya SSH se). IaaS example hai — aap OS aur upar sab manage karte ho. Use cases: web servers, app servers, development environments.

Q52. Azure VM sizes ke categories kya hain?

Azure VMs multiple families mein available hain:

- B-series: Burstable, general purpose, cost-effective
- D-series: Balanced CPU/Memory, most common
- E-series: Memory optimized (databases)
- F-series: Compute optimized (batch processing)
- N-series: GPU enabled (AI/ML, graphics)
- M-series: Massive memory (SAP HANA)

Har family ke multiple sizes hain jaise D2s_v3 (2 vCPU, 8GB RAM).

Q53. What is Azure Virtual Machine Scale Sets?

VM Scale Sets se aap ek saath multiple identical VMs deploy aur manage kar sakte ho. Automatic scaling support karta hai — CPU zyada ho toh VMs badhao, kam ho toh ghataao. Load balancer automatically traffic distribute karta hai. Use case: web applications, batch processing, container workloads jahan high availability chahiye.

Q54. Azure App Service kya hai?

Azure App Service ek PaaS service hai jo web applications, REST APIs aur mobile backends host karne ke liye hai. Aap sirf code deploy karo — OS, servers, scaling sab Azure handle karta hai. Support karta hai .NET, Java, Node.js, Python, PHP, Ruby. Auto-scale, custom domains, SSL certificates, deployment slots sab built-in. Developers ke liye perfect.

Q55. Azure App Service Plans kya hote hain?

App Service Plan define karta hai ki aapka App Service kitne resources par chale:

- Free (F1): Limited, no SLA, shared resources
- Shared (D1): Custom domain, shared
- Basic (B1-B3): Dedicated VMs, no auto-scale
- Standard (S1-S3): Auto-scale, staging slots, backups
- Premium (P1v3-P3v3): Enhanced performance, VNet
- Isolated (I1-I3): Dedicated environment, maximum security

Q56. Azure Functions kya hain?

Azure Functions ek serverless compute service hai — aap sirf code likhte ho, baaki sab Azure sambhalta hai. Functions events se trigger hoti hain: HTTP request, timer, message queue, file upload, database change, etc. Pay-per-execution model. Use cases: email bhejne ka automation, image resize karna, API endpoints, scheduled tasks. Micro-billing isliye cost-effective hai.

Q57. Azure Container Instances (ACI) kya hai?

ACI Azure mein containers run karne ka sabse fast aur simple tarika hai — koi VM manage nahi karna, koi orchestration nahi. Single container ya container group run karo in seconds. Per-second billing. Use cases: batch jobs, event-driven tasks, dev/test, simple applications. Kubernetes ki complexity avoid karna ho toh ACI perfect hai.

Q58. Azure Kubernetes Service (AKS) kya hai?

AKS Azure ka managed Kubernetes service hai. Kubernetes ek container orchestration platform hai. AKS mein Azure handle karta hai: control plane, upgrades, scaling, monitoring. Aap sirf application containers manage karo. Use cases: microservices, large-scale containerized applications. AKS free hai — sirf worker node VMs ka cost lagta hai.

Q59. Virtual Machines vs App Service vs Functions — kab kya use karein?

VM – Jab full OS control chahiye, existing apps migrate karne ho (lift-and-shift), legacy applications. Most control, most responsibility.

App Service – Web apps aur APIs ke liye, OS management nahi chahiye, auto-scale chahiye. Middle ground.

Functions – Short-lived, event-driven tasks, infrequent workloads, pay-per-use chahiye. Least management, most automated.

Q60. Azure Batch kya hai?

Azure Batch large-scale parallel aur HPC (High Performance Computing) jobs run karne ke liye hai. Sab ke sab resources automatically provision aur manage hote hain. Use cases: rendering, video transcoding, financial modelling, scientific simulations. Hundreds ya thousands of VMs ek saath chalo, kaam hone par band karo. Cost effective high-compute tasks ke liye.

Q61. What is Azure Logic Apps?

Logic Apps ek low-code/no-code workflow automation service hai. 400+ connectors hain: Office 365, Salesforce, Twitter, SQL, SAP, etc. Visual designer mein workflows banao — koi code nahi likhna. Example: Jab customer email aaye toh automatically CRM mein entry karo aur Slack notification bhejo. Business process automation ke liye perfect.

Q62. Azure DevOps kya provide karta hai?

Azure DevOps ek complete software development platform hai:

- Azure Repos: Git code repositories
- Azure Pipelines: CI/CD automation
- Azure Boards: Agile project tracking
- Azure Artifacts: Package management (npm, Maven, NuGet)
- Azure Test Plans: Test management

Ye sab together development teams ko code likhne se production deploy karne tak help karta hai.

Q63. What is Azure Virtual Desktop?

Azure Virtual Desktop (AVD) ek Desktop-as-a-Service (DaaS) hai. Windows 10/11 desktop cloud mein run hota hai, users access karte hain browser ya app se kisi bhi device se. Benefits: work from home, BYOD (Bring Your Own Device), centralized management, cost savings. Multi-session Windows support karta hai — multiple users ek VM share kar sakte hain.

Q64. Azure Spot VMs kya hain?

Spot VMs Azure ke unused compute capacity par run hote hain — up to 90% cheaper than regular VMs. Catch: Azure ko zaroorat hogi toh 30 second notice mein evict kar dega. Use cases: batch processing, non-critical workloads, dev/test, stateless applications. Price real-time mein fluctuate hoti hai. Agar eviction handle kar sako toh massive savings.

Q65. Reserved VM Instances kya hote hain?

Reserved Instances = 1 ya 3 saal ke liye specific VM size reserve karo, advance payment karo, 40-72% savings pao. Types:

- All Upfront: Max savings, ek baar pay karo.
- Partial Upfront: Mixed payment.
- No Upfront: Monthly pay karo, phir bhi ~40% savings.

Agar VM continuously chalani hai (production workloads), Reserved Instances best ROI dete hain.

Q66. What is Azure Compute Gallery?

Azure Compute Gallery (pehle Shared Image Gallery) ek centralized repository hai VM images manage karne ke liye. Custom images create karo, version control karo, globally replicate karo multiple regions mein. Teams images share kar sakti hain. Automation mein use hota hai — consistent VMs deploy karne ke liye same base image se.

Q67. Azure Container Registry kya hai?

Azure Container Registry (ACR) ek private Docker container image registry hai — DockerHub ka Azure equivalent. Teams apne custom container images yahan store karte hain securely. AKS, ACI, App Service direct ACR se images pull kar sakte hain. Geo-replication support karta hai. Role-based access control se security ensure hoti hai.

Q68. What is Azure Service Fabric?

Service Fabric ek platform hai microservices aur containers deploy, manage aur scale karne ke liye. Azure khud apni many internal services (Azure SQL, Cosmos DB, Skype) Service Fabric par run karta hai. Reliable services, stateful services, actor model support karta hai. Kubernetes se pehle ye popular tha; ab AKS preferred hai production ke liye.

Q69. Explain difference between containers and VMs.

VM – Full virtual computer. OS + Applications. GBs mein size. Minutes mein start. Strong isolation.

Container – OS share karte hain sirf app + dependencies. MBs mein size. Seconds mein start. Lightweight isolation.

Containers zyada efficient hain resources ke liye. VMs zyada isolated hain. Cloud native apps mostly containers use karti hain aaj. Azure dono support karta hai.

Q70. What is Azure Spring Apps?

Azure Spring Apps (pehle Azure Spring Cloud) ek PaaS service hai specifically Spring Boot (Java) microservices deploy karne ke liye. Microsoft aur VMware ne milke banaya. Developers sirf Java code likhein — infrastructure, scaling, monitoring, service discovery sab managed. Java developers ke liye highly optimized.

Q71. What is deployment slot in App Service?

Deployment Slots App Service ka feature hai — alag environments (staging, production) ek hi App Service plan mein. Pehle staging slot mein deploy karo, test karo, phir ek click mein production ke saath swap karo. Blue-green deployment possible. Agar kuch galat ho toh instantly swap back karo. Zero downtime deployments ke liye excellent.

Q72. What is Azure Red Hat OpenShift?

Azure Red Hat OpenShift (ARO) ek fully managed OpenShift (Kubernetes distribution by Red Hat) service hai Azure par. Microsoft aur Red Hat jointly manage karte hain. Enterprises jo already Red Hat tech use karti hain unke liye Azure migration easy hoti hai. Compliance, security aur support dono providers se milta hai.

Q73. What is Azure Dedicated Host?

Azure Dedicated Host ek physical server hai sirf aapke liye — kisi aur customer ke VMs uss server par nahi honge. Benefits: compliance requirements (single-tenant), maintenance control, visibility of physical hardware. Cost zyada hota hai. Use case: banking, government, healthcare jahan data isolation critical hai.

Q74. What is proximity placement groups in Azure?

Proximity Placement Groups ensure karte hain ki aapki multiple VMs ek hi data center mein physically close hon — network latency minimum ho. Use case: tightly coupled applications jahan inter-VM communication fast chahiye (HPC, SAP workloads, gaming backends). Tradeoff: high availability thodi compromise hoti hai.

Q75. Azure mein availability set kya hoti hai?

Availability Set ensure karta hai ki VMs alag fault domains aur update domains mein hon:

- Fault Domain: Alag power aur network — hardware failure se protection.
- Update Domain: Azure updates ek domain pe karta hai pehle, doosra running rehta hai.

Result: Multiple VMs ka group kabhi ek saath down nahi hoga. 99.95% SLA milti hai. Same region ke andar protection hai, cross-zone nahi.

Q76. What are Azure VM extensions?

VM Extensions small programs hain jo VM par deploy aur run hote hain — configuration, monitoring, security ke liye. Example:

- Custom Script Extension: Script run karo VM creation ke time.
- Azure Monitor Agent: Logs collect karo.
- Desired State Configuration (DSC): Ensure VM configured rahe correctly.

Portal, CLI, ARM templates se extensions manage karo.

Q77. What is Azure Autoscale?

Azure Autoscale automatically resources badhata-ghataata hai demand ke hisaab se:

- Scale Out: Zyada VMs/instances add karo.
- Scale In: Kam karo jab demand kam ho.

Rules set karo: 'Jab CPU > 80% ho 10 min tak, toh 2 instances add karo.'

Schedule-based bhi ho sakta hai. App Service, VM Scale Sets, AKS sab support karte hain autoscale.

Q78. What is Azure Load Balancer?

Azure Load Balancer incoming traffic distribute karta hai multiple VMs/instances mein — equal distribution ke liye. Types:

- Basic Load Balancer: Free, limited features.
- Standard Load Balancer: Production-grade, zone-redundant, HTTPS health probes.

Layer 4 (TCP/UDP) par kaam karta hai. Web applications mein traffic evenly distribute karne ke liye use hota hai.

Q79. What is Azure Application Gateway?

Application Gateway Layer 7 (HTTP/HTTPS) load balancer hai. Features:

- URL-based routing (different URLs → different backends)
- SSL termination
- Web Application Firewall (WAF) integration
- Cookie-based session affinity

Normal Load Balancer se zyada intelligent hai — web traffic ko samajhta hai. E-commerce, SaaS applications ke liye best.

Q80. What is Azure Traffic Manager?

Traffic Manager DNS-based global load balancer hai. Different regions mein deployed applications ke beech traffic route karta hai based on:

- Performance: Nearest/fastest region.
- Priority: Primary failover to secondary.
- Weighted: % basis par distribute.
- Geographic: User ki location ke basis par.

Ye actual data transfer nahi karta — sirf DNS response change karta hai.

Section 3: Azure Core Services – Storage

Q81. Azure Storage account kya hota hai?

Azure Storage Account ek container hai jo Azure ke storage services hold karta hai. Har storage service (Blob, Files, Queue, Table) ek account ke under hoti hai. Account name globally unique hona chahiye. Types: General Purpose v2 (recommended), Blob Storage, Premium. Access tiers, replication options account level par configure hote hain.

Q82. Azure Blob Storage kya hai?

Blob Storage unstructured data ke liye hai – images, videos, documents, backups, logs. 'Blob' = Binary Large Object. Ek storage account mein Containers hote hain (folders jaisi), unke andar Blobs (files). Types:

- Block Blob: Files ke liye (most common)
 - Append Blob: Log files ke liye
 - Page Blob: VM disk files (.vhd) ke liye
- Massive scale, low cost, globally accessible.

Q83. Blob Storage access tiers kya hote hain?

Azure Blob Storage mein 4 access tiers hain:

1. Hot – Frequently accessed data. High storage cost, low access cost.
 2. Cool – Infrequently accessed (30+ days). Lower storage cost, higher access cost.
 3. Cold – Rarely accessed (90+ days). Even lower storage cost.
 4. Archive – Long-term backup/compliance. Cheapest storage, but retrieval takes hours.
- Lifecycle policies se automatic tier transitions set kar sakte hain.

Q84. Azure Files kya hai?

Azure Files cloud-based file share hai jo SMB (Server Message Block) ya NFS protocol use karta hai. Windows/Linux/Mac machines se mount kar sako jaise network drive. Use cases: shared application config files, lift-and-shift file server migration, app needing shared file storage. Azure File Sync se on-premises file server ke saath sync karo.

Q85. Azure Queue Storage kya hai?

Queue Storage message queuing service hai – components ke beech messages store karta hai asynchronous communication ke liye. Ek queue mein millions of messages ho sakte hain (up to 64KB each). Use case: web application order submit karta hai queue mein, backend service process karta hai apni speed se. Decoupling architecture ke liye perfect. FIFO nahi necessarily – peek karo, process karo, delete karo.

Q86. Azure Table Storage kya hai?

Table Storage NoSQL key-value store hai structured data ke liye — schema-less. Large amounts of structured data sasta store karo. Use cases: user preferences, telemetry data, address books. Azure Cosmos DB Table API iska enhanced version hai. Cosmos DB zyada features deta hai, Table Storage zyada sasta aur simple hai.

Q87. Azure Disk Storage kya hai?

Azure Managed Disks VM ke liye persistent block storage hain:

- Ultra Disk: Highest performance (SAP HANA, top-tier DBs)
- Premium SSD: Production workloads
- Standard SSD: Web servers, dev/test
- Standard HDD: Backups, infrequent access

Managed Disks automatically replicated hoti hain — aapko storage account manage nahi karna. Azure availability sets ke saath automatically aligned hoti hain.

Q88. Storage redundancy options kya hain Azure mein?

LRS (Locally Redundant Storage) – 3 copies, same data center. Cheapest.

ZRS (Zone Redundant Storage) – 3 copies, 3 different availability zones. Same region.

GRS (Geo Redundant Storage) – LRS + 3 copies in secondary region. 6 copies total.

GZRS (Geo Zone Redundant Storage) – ZRS + 3 copies in secondary region. Maximum durability.

GRS/GZRS secondary region: read access only with RA-GRS/RA-GZRS option.

Q89. Azure Storage security features kya hain?

1. Encryption at rest – AES 256-bit, by default.
2. Encryption in transit – HTTPS enforced.
3. Shared Access Signature (SAS) – Limited time/permission access tokens.
4. Storage Account Keys – Full access keys (treat like passwords).
5. Azure AD authentication – Role-based access.
6. Private Endpoints – Storage account ko VNet mein privately access karo.
7. Soft Delete – Accidental deletion se protection.
8. Azure Defender for Storage – Threat detection.

Q90. What is Shared Access Signature (SAS)?

SAS ek security token hai jo limited access deta hai Azure Storage ko:

- Kaunse resources access ho saktein (specific blob, container, service)
- Kya permissions hain (read, write, delete, list)
- Kab tak valid hai (expiry time)

Use case: Customer ko temporarily ek file download karne do bina storage account key expose kiye. 3 types: Service SAS, Account SAS, User Delegation SAS (most secure — AD backed).

Q91. Azure Blob lifecycle management kya hai?

Lifecycle Management policies se aap automate kar sakte ho blob tier transitions aur deletion:

- 30 din baad Hot → Cool
- 90 din baad Cool → Cold
- 180 din baad → Archive
- 365 din baad → Delete

Cost optimization ke liye very useful. Logs aur backups jo time ke saath less important ho jaate hain unhe automatically cheap tier mein shift karo.

Q92. Azure Data Lake Storage Gen2 kya hai?

ADLS Gen2 = Azure Blob Storage + Hierarchical Namespace + Big Data analytics capabilities. Hadoop-compatible filesystem. Petabyte scale data store for analytics. Azure Databricks, Azure Synapse Analytics, HDInsight ke saath use hota hai. Data engineers ke liye raw data store karne ka preferred option. Hierarchical namespace se directories properly manage hoti hain.

Q93. Azure Storage Explorer kya hai?

Azure Storage Explorer ek free desktop application hai (Windows/Mac/Linux) jo graphically Azure Storage manage karne deta hai — portal ki jagah. Blobs, tables, queues, files sab ko GUI se manage karo. On-premises emulators se bhi connect karo (Azurite). Developers aur admins ke liye daily use tool hai testing aur management ke liye.

Q94. What is Azure File Sync?

Azure File Sync on-premises Windows file servers aur Azure Files ke beech sync karta hai. Cloud tiering feature: less-used files automatically Azure par move ho jaate hain, frequently accessed files local cache mein rehte hain — same drive space mein effectively unlimited storage. Branch offices ke liye centralized file management enable karta hai.

Q95. What is immutable storage in Azure?

Immutable Blob Storage ensure karta hai ki data ek specified time tak delete ya modify na ho sake — WORM (Write Once, Read Many). Types:

- Time-based retention: Specific duration tak protected.
- Legal hold: Explicitly released tab tak protected.

Use cases: Financial records, legal documents, compliance. FINRA, SEC regulations ke liye required. Azure Blob Storage mein built-in feature hai.

Q96. What is Azure Backup?

Azure Backup cloud-based backup service hai:

- Azure VMs backup karo
- On-premises servers/VMs backup karo
- SQL databases backup karo
- Azure Files shares backup karo

Centralized management Recovery Services vault se. LRS ya GRS redundancy. Soft delete protection — accidental deletion ke 14 din baad bhi recover kar sako. No infrastructure maintain karna.

Q97. What is Azure Site Recovery?

Azure Site Recovery (ASR) Disaster Recovery as a Service hai. On-premises ya other cloud VMs ko Azure mein replicate karo. Failover: disaster hone par Azure mein VMs start ho jaayein. Failback: Original site recover hone par wapas shift karo. Supports VMware, Hyper-V, physical servers. Regular DR drills kar sako production impact ke bina. RTO/RPO minimize karo.

Q98. Storage account ki performance tiers kya hain?

Standard Performance – HDDs ya cheaper SSDs, HDD-backed, lower cost, general purpose. Most use cases ke liye adequate.

Premium Performance – NVMe SSDs, ultra-low latency, higher cost. Use cases: High transaction workloads (Premium Block Blob, Premium File Shares, Premium Page Blobs).

Note: Premium storage accounts sirf LRS ya ZRS redundancy support karte hain — GRS nahi.

Q99. What is Azure Content Delivery Network (CDN)?

Azure CDN static content (images, CSS, JS, videos) globally cache karta hai — edge servers par. User request aata hai nearest edge server se content deliver hota hai, origin server tak nahi jaana padta. Isse latency drastically kam hoti hai aur origin server ka load kam hota hai. Azure CDN providers: Azure CDN from Microsoft, Verizon, Akamai.

Q100. Azure Storage pricing kya factors determine karte hain?

Azure Storage pricing ke main factors:

1. Data stored (per GB per month)
2. Operations (read/write/list per 10,000 transactions)
3. Data transfer out (egress charges)
4. Replication type (LRS sasta, GRS costly)
5. Access tier (Archive cheapest storage, expensive access)
6. Region (prices vary by geography)

Azure Pricing Calculator se accurate estimate lagao deploy karne se pehle.

Q101. What is the difference between Azure Files and Azure Blob Storage?

Azure Blob Storage – Object storage for unstructured data. Not mountable as file system. REST API access. Best for large files, media, backups.

Azure Files – Fully managed file shares. SMB/NFS mountable – Windows Explorer ki tarah use karo. Best for shared app configs, lift-and-shift file server scenarios.

Dono cases ke liye pricing alag hai. Typically Blob cheaper than Files for same data volume.

Q102. What is Azure NetApp Files?

Azure NetApp Files enterprise-grade high-performance NAS (Network Attached Storage) service hai – NetApp ke technology par based. Ultra-low latency (sub-millisecond), high throughput. Mission-critical workloads: Oracle databases, SAP, high-performance computing. Costlier than regular Azure Files but far more performant for demanding workloads.

Q103. What is Azurite?

Azurite ek local Azure Storage emulator hai (Azure Storage Emulator ka successor). Developers apne local machine par Azure Blob, Queue, Table storage simulate kar sakte hain bina actual Azure account ke. Node.js mein written, cross-platform. CI/CD pipelines mein test environments ke liye use hota hai. npm install -g azurite se install hota hai.

Q104. Storage account mein network access restrict kaise karein?

Storage account ko specific networks se restrict karo:

1. Firewalls & Virtual Networks – Specific VNets ya IP ranges allow karo.
2. Private Endpoints – VNet ke andar private IP deta hai storage account ko.
3. Service Endpoints – VNet se storage traffic directly route karo Azure backbone par.
4. Trusted Azure Services – Azure services bypass kar sakti hain firewall.

Default public internet access band karo production environments mein.

Q105. Azure Storage redundancy mein 11 nines durability kya matlab hai?

11 nines = 99.999999999% durability. Matlab agar aap 10 million objects store karo, toh expected loss har 10 million saalon mein 1 object hai. Ye practically zero data loss hai. GRS aur GZRS storage 16 nines durability provide karta hai. LRS bhi 11 nines deta hai. Yeh replication ke through multiple copies se achieve hota hai.

Q106. What is Azure Data Box?

Azure Data Box physical device hai large amounts of data Azure mein transfer karne ke liye – internet pe time lage ya bandwidth limited ho toh. Data Box ek 80TB rugged device hai jo aapke location bheja jaata hai, data copy karo, wapas bhejo, Microsoft Azure mein upload kar deta hai. Variants: Data Box Disk (8TB SSD), Data Box (80TB), Data Box Heavy (1PB).

Q107. Object replication in Azure Blob Storage kya hai?

Object Replication asynchronously block blobs copy karta hai source container se destination container mein – different storage accounts/regions mein. Use cases: latency minimize karo (nearest region mein copy), analytics workload optimization, backup/DR. GRS se alag – object replication user-defined rules follow karta hai, GRS automatic hai.

Q108. What is blob versioning in Azure?

Blob Versioning automatically previous versions of a blob save karta hai jab bhi modified ya deleted ho. Accidental overwrites aur deletions se protection. Each version ek unique ID rakhti hai. Previous versions restore karo anytime. Lifecycle policies se old versions automatically delete ho sakte hain to control cost. Compliance aur audit trails ke liye useful.

Q109. What is Azure Premium SSD v2?

Premium SSD v2 Azure ka most advanced managed disk tier hai (baad mein add hua). Sub-millisecond latency, high IOPS/throughput independently configure karo bina higher tier pick kiye. Cost-efficient for demanding workloads. Supports online resize aur performance changes without downtime. Ultra Disk se thoda kam extreme, Premium SSD v1 se kaafi zyada capable.

Q110. Summarize Azure Storage services for exam.

AZ-900 ke liye remember karo:

- Blob: Unstructured data, 3 types (Block/Append/Page), 4 access tiers
- Files: Mountable file shares, SMB/NFS
- Queue: Message queuing, async communication
- Table: NoSQL structured data
- Disk: VM storage (Managed Disks)
- Redundancy: LRS < ZRS < GRS < GZRS
- Security: Encryption default, SAS tokens, Azure AD, Private Endpoints

Section 4: Azure Core Services – Networking

Q111. Azure Virtual Network (VNet) kya hota hai?

VNet Azure mein private network hai — jaise aapke office ka local network, lekin cloud mein. VMs, databases aur services VNet ke andar private IPs se communicate karte hain. Internet se isolated rehta hai by default. Subnets se VNet divide kar sakte ho. Resource Group aur Region specific hoti hai. Foundation hai Azure networking ki.

Q112. Subnet kya hota hai?

Subnet VNet ka ek division hai — ek bade network ko logical segments mein todna. Benefits: traffic segregate karo (web tier, app tier, database tier alag subnets mein), NSGs apply karo per subnet, routing control karo. Example: VNet 10.0.0.0/16 □ Web subnet 10.0.1.0/24, App subnet 10.0.2.0/24, DB subnet 10.0.3.0/24.

Q113. Network Security Group (NSG) kya hai?

NSG ek virtual firewall hai jo Azure resources ke network traffic ko filter karta hai. Inbound aur outbound rules define karo — port, protocol, source/destination ke basis par allow ya deny. NSG subnet ya individual NIC par attach karo. Default rules: Deny all inbound internet, Allow all outbound internet. Priority number se rules order decide hota hai (100 = highest).

Q114. Azure VPN Gateway kya hai?

VPN Gateway on-premises network aur Azure VNet ke beech encrypted connection banata hai — internet ke through. Types:

- Site-to-Site (S2S): Office network □ Azure VNet
- Point-to-Site (P2S): Individual user's laptop □ Azure VNet
- VNet-to-VNet: Different VNets connect karna

IPsec/IKE encryption use karta hai. ExpressRoute se slow/cheap hai lekin simple setup.

Q115. Azure ExpressRoute kya hai?

ExpressRoute dedicated private connection hai aapke on-premises network aur Azure ke beech — public internet se nahi jaata. Benefits: Higher bandwidth, lower latency, consistent performance, better security. Telecom provider ke through setup hota hai. Expensive lekin mission-critical workloads ke liye ideal. 50Mbps se 100Gbps tak speeds available.

Q116. VNet Peering kya hota hai?

VNet Peering do VNets ko connect karta hai directly — bina VPN Gateway ya internet ke. Traffic Azure backbone par chalta hai (low latency, high bandwidth). Types:

- Regional Peering: Same region ke VNets.
- Global Peering: Different regions ke VNets.

Bi-directional traffic nahi hota by default — dono sides se peering configure karna padta hai. Hub-and-spoke architecture mein use hota hai.

Q117. Azure DNS kya hai?

Azure DNS domain names ko IP addresses mein resolve karta hai — Microsoft ke global DNS infrastructure par. Benefits: Azure ke saath integrated (same portal, same billing), SLA, geo-distribution.

- Public DNS Zones: internet-facing domains ke liye
- Private DNS Zones: VNet ke andar naam resolution ke liye

Azure DNS khud domain purchase nahi karta — Domain register karo kisi bhi registrar se, Azure mein host karo.

Q118. Azure Firewall kya hai?

Azure Firewall ek fully managed, cloud-native network security service hai. NSG se advanced features:

- FQDN (domain name) filtering
- Threat intelligence based filtering
- Outbound/inbound traffic filtering at scale
- DNAT rules (port forwarding)
- Application aur network rules

Hub-and-spoke network topology mein hub VNet mein deploy karo, sab traffic isse through karo. Premium tier mein IDS/IPS bhi hai.

Q119. What is Azure DDoS Protection?

DDoS (Distributed Denial of Service) Protection service Azure resources ko volumetric attacks se protect karti hai:

- Basic: Automatically enabled for free, platform-level protection.
- Standard: Enhanced protection, automatic tuning, cost guarantee (attack time billing nahi), real-time metrics aur alerts.

Standard plan mein Azure resources ka financial protection bhi hai — attack ke time autoscale se bade bill nahi aaye.

Q120. What is Azure Private Link?

Azure Private Link Azure services ko aapki VNet mein private IP pe accessible karta hai — public internet se nahi. Jaise Azure SQL, Storage, ya custom services private endpoint ke through access karo. Data Microsoft network par hi rehta hai. Compliance ke liye important. Private Endpoint ek NIC hoti hai VNet mein jo service se map hoti hai.

Q121. What is Azure Front Door?

Azure Front Door global HTTP/HTTPS load balancing, CDN aur WAF service hai. Features:

- Global routing (Anycast)
- SSL offload
- URL-based routing
- Caching
- WAF integration
- Health probes

Ye internet-facing applications ke liye best option hai — Application Gateway regional hai, Front Door global hai. E-commerce, gaming, media streaming ke liye perfect.

Q122. What is Azure Bastion?

Azure Bastion secure aur seamless RDP/SSH access deta hai VMs ko directly Azure portal se — bina public IP ke. Traditional approach: VM ko public IP do → attack surface badhta hai. Bastion approach: Bastion host VNet mein deploy karo, browser se connect karo — VM ko public IP nahi chahiye. Security best practice hai ye approach.

Q123. Route Tables kya hote hain Azure mein?

Route Tables custom network routes define karte hain — traffic kahan jaaye. Default Azure routes automatically kaam karte hain. Custom routes se:

- Traffic Network Virtual Appliance (firewall) ke through route karo
- Internet-bound traffic on-premises redirect karo
- VNet ke beech specific paths define karo

UDR (User Defined Routes) bhi kehte hain inhe. Each subnet ek route table se associate kar sakte ho.

Q124. What is Network Virtual Appliance (NVA)?

NVA Azure Marketplace se available third-party network devices hain — firewall, load balancer, WAN optimizer (Palo Alto, Fortinet, Check Point, F5, etc.). Custom routing ke saath VNet mein deploy karo. Azure Firewall Microsoft ka managed NVA hai. Enterprises jo on-premises specific vendor use karte hain, same vendor ka NVA Azure mein deploy kar sakte hain.

Q125. What is Service Endpoint in Azure networking?

Service Endpoints VNet se directly Azure services (Storage, SQL, KeyVault, etc.) tak optimized route provide karte hain — traffic public internet par nahi jaata, Azure backbone par jaata hai. Resources ko VNet specific access restrict karo. Private Endpoint se alag: Service Endpoint traffic VNet se public service endpoint tak jaata hai; Private Endpoint service ko private IP deta hai.

Q126. Azure Network Watcher kya hai?

Network Watcher Azure network monitoring aur diagnostics tool hai:

- IP Flow Verify: Check karo traffic allow/deny kyun ho raha hai
- Next Hop: Packet ka next hop kya hoga
- Connection Troubleshoot: VM se endpoint connectivity test karo
- Packet Capture: Network traffic capture karo
- Topology: Visual network diagram
- NSG Flow Logs: Traffic logs analyze karo

Network issues troubleshoot karne ke liye essential tool.

Q127. Hub-and-Spoke network topology kya hai?

Hub-and-Spoke ek common Azure network design pattern hai:

- Hub VNet: Central VNet jisme shared services hain (Firewall, VPN, Bastion, DNS).
- Spoke VNets: Individual application VNets jo hub se peered hain.

Benefits: Centralized security, shared connectivity services, isolation. Ek Azure Firewall se sab spokes ka traffic inspect karo. Enterprise-scale networking ke liye recommended pattern.

Q128. What is Azure Virtual WAN?

Azure Virtual WAN (vWAN) ek networking service hai jo hub-and-spoke, branch connectivity, aur global transit networking automate karta hai. Multiple branch offices, on-premises data centers ko Azure aur ek doosre se connect karo easily. SD-WAN vendors se integrate hota hai. Large enterprises ke liye manual VPN/peering setup replace karta hai.

Q129. What are Azure networking best practices?

1. NSGs har subnet par apply karo.
2. VMs ko public IP mat do — Azure Bastion use karo.
3. Least privilege network access.
4. DDoS Standard enable karo production par.
5. Network Watcher enable karo diagnostics ke liye.
6. Hub-and-spoke topology use karo enterprises ke liye.
7. Private Endpoints use karo sensitive services ke liye.
8. Azure Firewall ya NVA egress traffic filter karne ke liye.

Q130. What is the difference between Azure Load Balancer and Application Gateway?

Azure Load Balancer – Layer 4 (TCP/UDP). Fast, simple, no content awareness. Backend pool = VMs/scale sets. Internal ya public.

Application Gateway – Layer 7 (HTTP/HTTPS). URL routing, SSL termination, WAF, session affinity, redirects. Understands web traffic.

Kab use karein: Pure TCP/UDP load balancing □ Load Balancer. Web app intelligent routing □ Application Gateway. Global HTTP(S) routing □ Azure Front Door.

Q131. What is Azure CDN integration with Storage?

Azure CDN ke saath Blob Storage integrate karke static website content globally fast deliver karo:

1. Storage account mein static website enable karo.
2. CDN endpoint create karo origin = storage account.
3. DNS configure karo custom domain ke liye.

Result: images, JS, CSS sab edge nodes se serve hote hain — origin server par load nahi. Cost bhi kam hoti hai kyunki storage egress CDN se sasta hota hai.

Q132. What is Azure Private DNS Zone?

Private DNS Zone VNet ke andar naam resolution provide karta hai — internet par expose nahi hota. Example: myapp.internal □ 10.0.0.4. VMs ek doosre ko naam se call kar sakti hain IP yaad kiye bina. Private Endpoints ke saath automatically DNS records create hote hain. Multiple VNets ko ek private zone se link kar sakte ho.

Q133. IP addressing in Azure VNet kaise kaam karta hai?

VNet create karte time address space define karo (e.g., 10.0.0.0/16). Subnets us range se smaller blocks lete hain. Azure 5 IPs reserve karta hai har subnet mein (first 4 + last 1). VMs ko Private IPs milti hain automatically (DHCP se). Public IPs optional hain — internet connectivity ke liye. Static ya Dynamic public IPs assign kar sakte ho.

Q134. What is Azure NAT Gateway?

NAT (Network Address Translation) Gateway subnet ke VMs ko outbound internet connectivity deta hai bina individual public IPs ke. Ek NAT Gateway multiple VMs share karte hain. Predictable static public IP addresses outbound ke liye. SNAT exhaustion avoid karta hai (large scale deployments mein common problem). Outbound connections ke liye recommended approach.

Q135. What is Azure Private Peering in ExpressRoute?

ExpressRoute mein 3 peering types hain:

- Private Peering: Azure VNets se connect karo privately.
- Microsoft Peering: Microsoft 365, Dynamics 365 services se connect karo.
- Public Peering: (Deprecated) Azure public services access karna tha.

Private Peering se VMs Azure mein directly on-premises se accessible hoti hain — sirf private IPs, no internet.

Q136. What is Azure Network Performance Monitor?

Network Performance Monitor (NPM) ek Azure Monitor solution hai jo network connectivity aur performance monitor karta hai:

- On-premises ↔ Azure connectivity
- Azure ↔ Internet paths
- ExpressRoute monitoring
- Service connectivity (specific endpoints tak latency/loss)

Proactively issues detect karo degradation hone se pehle alert receive karo. Hybrid networks ke liye key monitoring tool.

Q137. What is the purpose of Azure Firewall Premium?

Azure Firewall Premium mein additional features hain:

- IDPS (Intrusion Detection and Prevention System) – Known threats signature-based detection.
- TLS Inspection – Encrypted traffic bhi inspect karo.
- URL Filtering – Specific URLs block karo.
- Web Categories – Categories of websites allow/block karo.

Use cases: PCI DSS compliance, advanced threat protection needed, heavily regulated industries.

Q138. What is Azure Virtual Network Gateway?

Virtual Network Gateway VNet ke liye gateway hai VPN ya ExpressRoute connections ke liye. VPN Gateway (IPsec/IKE) ya ExpressRoute Gateway ho sakta hai. SKUs: Basic, VpnGw1-5, ErGw1-3 — bandwidth aur features vary karte hain. VPN Gateway ek subnet mein deploy hoti hai jis subnet naam hona chahiye 'GatewaySubnet'. Azure ka requirement hai ye.

Q139. What is cross-premises connectivity in Azure?

Cross-premises = on-premises network aur Azure ke beech connectivity. Options:

1. VPN Gateway (Site-to-Site): Internet over encrypted tunnel. Budget option.
2. ExpressRoute: Private dedicated connection. Enterprise option.
3. ExpressRoute + VPN: Failover scenario — ExpressRoute primary, VPN backup.
4. Azure Virtual WAN: Large scale multi-site connectivity.

Hybrid cloud architecture ke liye fundamental building block.

Q140. Summarize Azure Networking for AZ-900.

Key points:

- VNet: Private network foundation
- Subnet: VNet ka division
- NSG: Virtual firewall (port/protocol rules)
- VPN Gateway: Encrypted internet-based hybrid connectivity
- ExpressRoute: Private dedicated hybrid connectivity
- Load Balancer: Layer 4 TCP/UDP balancing
- Application Gateway: Layer 7 web balancing + WAF
- Front Door: Global HTTP routing + CDN
- Bastion: Secure VM access without public IP
- DDoS Protection: Attack mitigation

Section 5: Azure Identity, Access & Security

Q141. Azure Active Directory (Azure AD) kya hai?

Azure AD (ab Microsoft Entra ID) Microsoft ka cloud-based identity aur access management service hai. Traditional Active Directory on-premises servers ke liye tha. Azure AD cloud ke liye hai – web apps, SaaS apps, Microsoft 365, Azure resources sab ek jagah se manage karo. Users, groups, applications, devices manage hote hain ismen. Free tier available hai.

Q142. Azure AD vs Traditional Active Directory (AD DS) mein kya farak hai?

Traditional AD (AD DS) – On-premises, Kerberos/NTLM protocols, OUs hierarchy, Group Policy, domain join karna hota hai.

Azure AD – Cloud-based, OAuth 2.0/OIDC/SAML protocols, flat structure, Conditional Access policies, device registration.

Azure AD Connect se dono sync ho sakte hain – hybrid identity. Aaj mostly Azure AD Hybrid join ya Azure AD join preferred hai modern environments mein.

Q143. Azure AD Free vs Premium P1 vs P2 kya hai?

Free – Basic user/group management, SSO (10 apps), MFA.

Premium P1 – Conditional Access, hybrid identity, dynamic groups, self-service password reset for on-premises, Microsoft Entra ID Protection basic.

Premium P2 – Identity Protection (risk-based Conditional Access), Privileged Identity Management (PIM), Access Reviews.

Microsoft 365 E3 mein P1, E5 mein P2 included hoti hai.

Q144. Single Sign-On (SSO) kya hota hai?

SSO = ek baar login karo, sab apps access karo. Example: Office 365 mein login kiya toh Salesforce, Box, ServiceNow sab automatically logged in. Benefits: Ek password yaad rakhna, less password fatigue, IT kam password resets handle kare, better security kyunki users strong password ek jagah rakhte hain. Azure AD 1000+ pre-integrated SaaS apps support karta hai.

Q145. Multi-Factor Authentication (MFA) kya hai?

MFA 2 ya zyada factors se identity verify karta hai:

1. Something you know: Password
2. Something you have: Phone (Authenticator app, SMS)
3. Something you are: Fingerprint, Face ID

Agar password leak bhi ho, MFA ke bina attacker login nahi kar sakta. Azure AD mein MFA free tier mein bhi available hai. Microsoft Authenticator app recommended method hai. Security default mein MFA automatically enable hota hai.

Q146. Conditional Access kya hai?

Conditional Access = IF (condition) THEN (action). Example:

- IF user is accessing from outside India AND is admin THEN require MFA.
- IF device is not compliant THEN block access.
- IF risky sign-in detected THEN force password reset.

Conditions: Location, device state, user risk, app, sign-in risk. Actions: Allow, Block, Require MFA, Require compliant device. Azure AD Premium P1/P2 feature hai.

Q147. Role-Based Access Control (RBAC) kya hai Azure mein?

RBAC users ko sirf wahi permissions deta hai jo unhe chahiye — principle of least privilege. 3 components:

- Security Principal: User, Group, Service Principal, Managed Identity
 - Role Definition: Permissions ka set (Owner, Contributor, Reader, ya custom)
 - Scope: Management Group, Subscription, Resource Group, ya individual Resource
- Role assignment = Principal + Role + Scope. Inherited hoti hai hierarchically.

Q148. Azure built-in RBAC roles kaunsi hain?

Common built-in roles:

- Owner: Full access + assign roles
- Contributor: Full access lekin roles assign nahi kar sakta
- Reader: Sirf dekh sakta hai
- User Access Administrator: RBAC manage karo
- Virtual Machine Contributor: VM manage karo lekin VNet/Storage nahi
- Storage Blob Data Reader/Contributor: Blob data access
- Network Contributor: Network resources manage karo

300+ built-in roles hain Azure mein. Custom roles bhi banao.

Q149. Managed Identity kya hoti hai?

Managed Identity Azure resources ko automatically Azure AD identity deta hai — application code mein credentials store nahi karne. Types:

- System-assigned: Resource ke saath create/delete hoti hai. 1:1 relationship.
- User-assigned: Independently create karo, multiple resources share kar sakti hain.

Example: VM ko Azure Key Vault se secrets read karne do bina password hardcode kiye. Best practice hai service-to-service authentication ke liye.

Q150. Service Principal kya hota hai?

Service Principal ek Azure AD identity hai applications, services ya automation tools ke liye — individual users ke liye nahi. Jaise application ka 'user account'. Azure DevOps pipelines, Terraform, scripts Azure resources access karne ke liye service principal use karte hain. Client ID + Client Secret ya Certificate se authenticate hota hai. Managed Identity preferred hai lekin SP bhi widely used hai.

Q151. Azure Key Vault kya hai?

Azure Key Vault secrets, keys aur certificates securely store karne ka service hai:

- Secrets: Passwords, connection strings, API keys
- Keys: Encryption keys (HSM-backed bhi)
- Certificates: SSL/TLS certificates

Applications Key Vault se runtime par secrets retrieve karein — code mein hardcode mat karo. Access policies ya RBAC se control karo kaun kya dekh sake. Soft delete + purge protection se accidental deletion rokein.

Q152. Azure Defender / Microsoft Defender for Cloud kya hai?

Microsoft Defender for Cloud (pehle Azure Security Center + Azure Defender) ek unified security management platform hai:

- Secure Score: Security posture ka score
- Recommendations: Security improve karne ke suggestions
- Alerts: Threat detection notifications
- Regulatory Compliance: ISO, PCI DSS, NIST status

Free tier mein basic recommendations. Defender plans enable karo specific resources ke liye advanced protection (VMs, Storage, SQL, etc.).

Q153. Azure Secure Score kya hota hai?

Secure Score aapki security posture ka numerical score hai (0-100%). Zyada score = better security. Microsoft Defender for Cloud mein available. Recommendations follow karo — jaise MFA enable karo, unnecessary ports band karo — score badhta hai. Executive dashboard ki tarah kaam karta hai — CISOs ko at-a-glance security health dikhata hai. Action items prioritize karne mein help karta hai.

Q154. Zero Trust security model kya hai?

Zero Trust = 'Never trust, always verify.' Traditional model: VPN ke andar sab trusted. Zero Trust model: Koi bhi trust nahi by default — verify karo explicitly.

Principles:

1. Verify explicitly: Always authenticate/authorize based on all available signals.
2. Use least privilege access: JIT, JEA, risk-based policies.
3. Assume breach: Minimize blast radius, segment access, encrypt everything.

Azure mein: Conditional Access, PIM, Entra ID Protection sab Zero Trust implement karte hain.

Q155. Privileged Identity Management (PIM) kya hai?

PIM (Azure AD P2 feature) privileged roles manage karta hai:

- Just-in-Time (JIT) access: Admin role sirf zaroorat hone par activate karo (1-8 hours).
- Approval workflow: Manager approve kare pehle.
- MFA required: Role activate karne pe MFA.
- Access reviews: Regular review ki ye log still need this role?
- Audit trail: Kab kab role activate hua track karo.

Breach ka risk kam hota hai kyunki permanent admin roles nahi hote.

Q156. Microsoft Entra ID kya hai?

Microsoft Entra ID = Azure Active Directory ka new name (2023 mein rename hua). Confusion avoid karne ke liye Microsoft ne apni identity products ko 'Microsoft Entra' family mein rebrand kiya. Features same hain — cloud identity, SSO, MFA, Conditional Access, etc. Exam mein dono names use ho sakte hain — samjho ki dono ek hi service hain.

Q157. What is Azure AD B2C?

Azure AD B2C (Business-to-Consumer) customer identity management ke liye hai — aapki apps mein customer login functionality. Features: Social login (Google, Facebook, Apple), custom branded login pages, MFA, self-service password reset. Millions of customers scale kar sakta hai. Developers apne mobile/web apps mein authentication Azure AD B2C ke through add karte hain.

Q158. What is Azure AD B2B?

Azure AD B2B (Business-to-Business) partner/vendor access management ke liye hai. External users (vendors, partners) ko aapki Azure resources access karne do unke existing identities se — naye accounts create karna nahi padta. Guest access dete hain. External users apne organization ka login use karte hain, aap apni resources par control rakhte ho. Collaboration use case.

Q159. What is Identity Protection in Azure AD?

Azure AD Identity Protection (P2 feature) machine learning se risky users aur risky sign-ins detect karta hai:

- Sign-in risk: Impossible travel, anonymous IP, unfamiliar location
- User risk: Leaked credentials detected

Policies set karo: Medium/High risk □ require MFA ya block.

Automatic response to compromised accounts without manual intervention. Security Operations teams ko alerts provide karta hai investigation ke liye.

Q160. What are Azure AD authentication methods?

Azure AD mein multiple authentication methods support hote hain:

1. Password hash sync (PHS): Hashed passwords sync hote hain Azure AD mein.
 2. Pass-through authentication (PTA): Password on-premises agent se validate hota hai.
 3. Federation (AD FS): Federated identity provider use karo.
 4. Passwordless: Windows Hello, FIDO2 keys, Microsoft Authenticator app.
- Passwordless most secure hai — password hi nahi toh leak bhi nahi hoga.

Q161. What is Microsoft Sentinel?

Microsoft Sentinel ek cloud-native SIEM (Security Information and Event Management) + SOAR (Security Orchestration Automated Response) solution hai. Large amounts of security data collect karo, AI se threats detect karo, automated responses configure karo. Azure, M365, third-party sources se logs integrate karo. Built-in MITRE ATT&CK; framework mapping. Large organizations ke security operations centers (SOC) ke liye.

Q162. What is Azure Policy?

Azure Policy governance tool hai jo ensure karta hai Azure resources compliance mein hain. Rules define karo:

- 'VMs sirf specific regions mein create ho sakti hain'
- 'Storage accounts HTTPS enforce karein'
- 'Tags mandatory hain'

Policies evaluate karte hain existing resources ko aur new deployments ko. Effect: Deny (block), Audit (report only), Append (tags add karo), DeployIfNotExists. Initiatives = multiple policies group.

Q163. What is Azure Blueprints?

Azure Blueprints repeatable set of Azure resources deploy karne ke liye template hai — governance package. Bundle karo: RBAC assignments, Azure Policies, ARM templates, Resource Groups. Ek new subscription setup karte waqt blueprint assign karo — sab automatically configured. Example: 'Secure Baseline Blueprint' = standard policies + roles apply karo instantly. Subscription-level governance.

Q164. What is Microsoft Defender for Identity?

Microsoft Defender for Identity (pehle Azure ATP) on-premises Active Directory threats detect karta hai. Domain Controllers par sensor install hota hai — AD traffic analyze karta hai, suspicious activity detect karta hai:

- Pass-the-hash attacks
- Lateral movement
- Privilege escalation
- Reconnaissance

Hybrid identity environments ke liye key security tool. Signals Microsoft Sentinel mein feed karta hai.

Q165. What is Azure AD Application Proxy?

Application Proxy on-premises web applications ko internet par securely publish karta hai — VPN ke bina. Kaise kaam karta hai: Connector on-premises install hota hai, Azure AD mein app register karo, external users Azure AD se authenticate ho ke on-premises app access karo. Zero trust remote access ke liye. Legacy apps jo Azure AD natively support nahi karti unhe bhi modernize karo.

Q166. What is RBAC vs Azure Policy difference?

RBAC — WHO kar sakta hai WHAT. Users/groups ko actions perform karne ki permissions. Azure Policy — WHAT hona chahiye. Resources compliant honi chahiye specific rules ke saath. Example:

- RBAC: 'Alice ko VMs create karne ka permission hai.'
- Policy: 'VMs sirf East US region mein create ho sakti hain.'

Dono milke kaam karte hain. RBAC access control ke liye, Policy governance ke liye.

Q167. What is Azure Information Protection?

Azure Information Protection (AIP) data classification aur protection ke liye hai — documents aur emails:

- Classify: Labels lagao (Public, Confidential, Highly Confidential)
- Protect: Encryption, permissions, visual markings
- Track: Document access track karo

Microsoft Purview Information Protection ke naam se rebrand hua. Sensitive data identify karo, automatically labels lagao, protection ensure karo wherever data goes — email karo, share karo, download karo.

Q168. What is Microsoft Purview?

Microsoft Purview data governance, risk aur compliance ke liye unified platform hai:

- Data Map: Discover aur catalog data assets
- Data Estate Insights: Data landscape visibility
- Information Protection: Classify aur protect data
- Data Loss Prevention (DLP): Sensitive data bahar na jaaye
- Compliance Manager: Regulatory compliance track karo
- Audit: Activity logs

CISOs aur data governance teams ke liye comprehensive tool.

Q169. What is Azure Dedicated HSM?

HSM = Hardware Security Module. Azure Dedicated HSM physical cryptographic hardware provide karta hai sirf aapke liye – cloud mein. Highest level of key protection. FIPS 140-2 Level 3 certified. Use cases: Payment processing, PKI management, banking applications. Regular Key Vault se alag – yahan keys ka lifecycle aap fully control karte ho, Microsoft bhi access nahi kar sakta.

Q170. What is compliance manager in Microsoft Purview?

Compliance Manager aapko regulatory compliance track aur manage karne mein help karta hai:

- Pre-built assessments: GDPR, ISO 27001, HIPAA, SOC2, etc.
- Compliance score: 0-100 percentage
- Improvement actions: Specific steps le ke score badhaao
- Control mapping: Apne controls ko regulatory controls se map karo

Compliance teams use karte hain ye ensure karne ke liye ki organization regulations follow kar rahi hai.

Q171. What is Azure Advisor security recommendations?

Azure Advisor aapko personalized recommendations deta hai 5 areas mein – Security bhi ek hai. Security recommendations Defender for Cloud se aati hain – integrate hain. Examples:

- 'Enable MFA for subscription owners'
- 'Remove deprecated accounts'
- 'Enable vulnerability assessment on VMs'

Free service hai, dashboard mein regularly check karo. High priority issues pehle fix karo.

Q172. What is resource lock in Azure?

Resource Locks accidentally delete ya modify karne se rokta hai:

- ReadOnly: Koi changes nahi kar sakta (sirf read)
- Delete: Delete nahi kar sakta, modify kar sakta hai

Lock management group, subscription, resource group ya individual resource par apply karo — inherited hota hai.

Example: Production database par Delete lock lagao — koi accidentally delete na kare. Owner role bhi delete nahi kar sakta — pehle lock remove karna padega.

Q173. What is Microsoft Cloud Security Benchmark?

Microsoft Cloud Security Benchmark (MCSB) Azure ke liye security best practices collection hai — CIS, NIST, PCI DSS se inspired. Categories: Network security, Identity management, Privileged access, Data protection, etc. Defender for Cloud automatically MCSB ke against evaluate karta hai aur recommendations deta hai. Security baseline set karne ke liye reference document hai teams ke liye.

Q174. What is Defense in Depth strategy?

Defense in Depth = multiple layers of security. Ek layer fail hone par doosra protect kare:

1. Physical: Data center security
2. Identity: MFA, Conditional Access
3. Perimeter: DDoS, Firewall
4. Network: NSGs, VNet segmentation
5. Compute: VM patches, endpoint protection
6. Application: Secure coding, OWASP
7. Data: Encryption, key management

Azure sab layers par security tools provide karta hai. Combination sabse effective hai.

Q175. What is Microsoft Entra Verified ID?

Microsoft Entra Verified ID decentralized identity solution hai — individuals aur organizations verified credentials issue aur verify kar sakte hain. Based on open standards (DID, Verifiable Credentials W3C standards). Examples: University degree credential issue karo, employer verify kare. No central database — users control apni identity. Privacy-preserving approach hai — minimum data share karo verification ke liye.

Q176. What is Azure Security Benchmark?

Azure Security Benchmark (ASB) Microsoft ka comprehensive set of security recommendations hai specifically Azure ke liye. Controls cover karte hain: network security, identity management, privileged access, data protection, asset management, logging, incident response, posture vulnerability management. Defender for Cloud ASB ke against score karta hai aapka. CIS Controls aur NIST CSF se mapped hai.

Q177. What is just-in-time (JIT) VM access?

JIT VM Access Defender for Cloud ka feature hai jo VM ports (RDP/SSH) sirf zaroorat pe kholti hai:

1. User request karta hai access (time limit specify karo)
2. Approval hone par NSG rule temporarily add hoti hai
3. Time expire hone par port automatically band ho jaata hai

Benefits: Brute force attacks ki window minimize hoti hai, audit trail hota hai, default mein ports closed rehte hain. Production VMs ke liye best practice.

Q178. What is the principle of least privilege?

Least Privilege = Sirf utna access do jitna zaroor ho — kuch nahi zyada. Examples:

- Database admin ko VM access nahi chahiye.
- Read-only user ko write permission nahi chahiye.
- Application ko sirf apna database access karne do.

Azure mein implement karo: RBAC se granular roles, PIM se JIT admin access, Managed Identities se app credentials, Conditional Access se context-aware access.

Breach ka impact minimize hota hai least privilege se.

Q179. What are Azure security tools summary?

AZ-900 ke liye key security tools:

- Microsoft Entra ID: Identity management, SSO, MFA
- Azure RBAC: Permission control
- Azure Policy: Governance, compliance
- Key Vault: Secrets, keys, certificates
- Defender for Cloud: Security posture, threat protection
- Sentinel: SIEM/SOAR, security analytics
- DDoS Protection: Attack mitigation
- Azure Firewall: Network security
- Bastion: Secure VM access
- PIM: Privileged identity management

Q180. What is identity as the new security perimeter?

Traditional security = Network perimeter (firewall ke andar = safe). Modern reality = Users work from anywhere, apps in cloud — network perimeter gone. New perimeter = Identity. Isliye Azure mein itna focus hai Azure AD (Entra ID) par:

- Strong authentication (MFA)
- Risk-based access (Identity Protection)
- Conditional Access policies
- Privileged Identity Management

Jis kisi ki identity compromise ho, uska access detect karo aur block karo quickly.

Section 6: Azure Cost Management & SLAs

Q181. Azure pricing ke main factors kya hain?

Azure pricing in factors se determine hoti hai:

1. Resource type: VM, Storage, Database, etc.
2. Service tier: Basic vs Standard vs Premium
3. Region: West US sasta, Brazil South costly
4. Bandwidth: Data transfer IN free, OUT charged
5. Support plan: Developer, Standard, Professional Direct
6. Licensing: Azure Hybrid Benefit se savings
7. Reservations: 1-3 year commit se 40-70% savings

Pricing Calculator se exact estimate lagao.

Q182. Azure Pricing Calculator kya hai?

Azure Pricing Calculator (azure.microsoft.com/pricing/calculator) ek free online tool hai jahan aap services select karo, configuration choose karo aur estimated monthly cost dekho – deploy karne se pehle. Multiple scenarios compare kar sakte ho. Configuration export karo share karne ke liye. Budget planning aur cost estimation ke liye essential tool.

Q183. Azure TCO Calculator kya hai?

TCO (Total Cost of Ownership) Calculator on-premises vs Azure ka comparison deta hai over 3-5 years. Aap enter karte ho:

- Current servers (count, specs)
- Storage
- Networking
- IT labor costs
- Software licenses

Calculator dikhata hai Azure mein shift karne par kitni savings honggi. Typically 30-70% savings dikhata hai. Sales/procurement conversations ke liye useful tool.

Q184. Azure Cost Management tool kya karta hai?

Azure Cost Management + Billing tool:

- Cost analysis: Resources per spending dekho
 - Budget alerts: Threshold cross hone par email
 - Advisor recommendations: Cost optimization
 - Cost allocation: Tags se department-wise chargeback
 - Invoice management: Download invoices
 - Exports: Data CSV mein export karo Power BI analysis ke liye
- Free tool hai, Azure portal mein available. FinOps teams ke liye core tool.

Q185. Azure Reserved Instances kaise savings deti hain?

Reserved Instances (RI) = 1 ya 3 saal ke liye specific VM/service commit karo:

- 1-year commitment: ~40% savings vs pay-as-you-go
- 3-year commitment: ~60-72% savings

Flexibility: Size flexibility (same family mein change), region flexibility, exchange (ek baar allowed), cancellation (with penalty).

Scope: Single subscription ya shared (multiple subscriptions). Production workloads jo continuously run hain unke liye perfect investment.

Q186. Azure Hybrid Benefit kya hai?

Azure Hybrid Benefit existing on-premises licenses use karke Azure mein savings karta hai:

- Windows Server with Software Assurance: Azure VM Windows license ka cost nahi lagta
- SQL Server with SA: Azure SQL/SQL on VM mein significant savings

For example, SQL Enterprise license Azure SQL mein use karo toh 80% savings ho sakti hain.

Organizations jo pehle se Microsoft licenses khareed chuke hain unke liye major cost optimization tool.

Q187. Azure Spot Instances ka use kab appropriate hai?

Spot Instances (up to 90% cheaper) appropriate hain:

- Batch processing jobs
- CI/CD testing pipelines
- Rendering workloads
- Big data analytics
- Development environments

NOT appropriate for:

- Production databases
- Active user-facing applications
- Stateful workloads without checkpointing

Eviction handling implement karo — state save karo, gracefully shutdown karo. Interruption-tolerant architectures ke liye best fit.

Q188. Azure Free Tier mein kya include hota hai?

Azure Free Account mein:

- \$200 credit: First 30 days ke liye
- 12 months free: Select popular services (750 hrs B1S Linux VM, 250GB SQL, 5GB Blob, 15GB bandwidth, etc.)
- Always free: 40+ services hamesha free:
- Azure Functions: 1M executions/month
- Azure Cosmos DB: 1000 RU/s, 25GB
- Azure App Service: 10 apps
- Azure DevOps: 5 users
- Azure Monitor: Basic logging

Students ke liye Azure for Students (\$100 credit) bhi hai.

Q189. Cost tagging kya hota hai Azure mein?

Tags key-value pairs hain resources par — metadata. Example: Department=Marketing, Environment=Production, CostCenter=1234.

Benefits:

- Cost allocation: Department wise cost dekho
- Chargeback: Teams ko unka bill dikhao
- Filtering: Reports filter karo
- Automation: Tagged resources differently treat karo

Azure Policy se mandatory tags enforce karo. Azure Cost Management mein tags se filter karo reports.

Best practice: Standard tagging taxonomy define karo organization ke liye.

Q190. Azure mein data transfer costs kaise kaam karti hain?

Data Transfer (Bandwidth) pricing:

- Inbound (data Azure mein aana): FREE
- Outbound (data Azure se bahar jaana): Charged per GB
- Same region ke andar: Mostly free ya very low
- Region to region (same geography): Low charge
- Cross-geography (e.g., US to Europe): Higher charge
- CDN se: Cheaper than direct storage egress

High data transfer applications ke liye architecture design karte waqt egress costs consider karo.

Q191. Azure support plans kya hain?

Azure Support Plans:

- Basic: Free. Documentation, forums, billing support only.
- Developer: \$29/month. Business hours email support. Dev/test environments.
- Standard: \$100/month. 24x7 phone+email. Production workloads.
- Professional Direct: \$1000/month. Faster response, proactive guidance, support engineer reviews.
- Premier/Unified: Enterprise level, dedicated account team.

SLA response times vary: Critical issue Standard = 2 hrs, Professional Direct = 1 hr.

Q192. What is an Azure SLA (Service Level Agreement)?

SLA Microsoft ka formal uptime/connectivity commitment hai for Azure services. Key points:

- Expressed as percentage uptime (e.g., 99.9%, 99.95%, 99.99%)
- Downtime allowed per year calculations:
99.9% = ~8.7 hours/year
99.95% = ~4.4 hours/year
99.99% = ~52 minutes/year
- SLA miss hone par: Service credits (billing discount)
- Free/Preview services: No SLA

AZ-900 mein composite SLA concept important hai.

Q193. Azure Cost Management alerts kaise set karte hain?

Cost Management mein Budget create karo:

1. Portal □ Cost Management □ Budgets □ Add
2. Scope: Subscription/Resource Group
3. Amount: Monthly budget set karo
4. Alert conditions: 80%, 100%, forecast 110% par alerts
5. Alert recipients: Email addresses

Result: Jab spending threshold cross ho, automatically email alert aayega. Surprises avoid karo. Multiple budgets set kar sakte ho different scopes ke liye.

Q194. Azure mein over-provisioning problem kya hai?

Over-provisioning = zyada resources lena jo use nahi ho rahe — waste of money.

Example: Production ke liye 16 CPU VM li, actual use 2 CPU hai.

Solutions:

1. Azure Advisor: Right-sizing recommendations deta hai
2. Azure Monitor: Actual usage dekho
3. Auto-scale: Need ke hisaab se scale karo
4. Reserved Instances sirf production ke liye
5. Regular cost reviews karo

Cloud mein over-provisioning thoda easy ho jaata hai — ease of provisioning se. FinOps culture important hai.

Q195. What is FinOps?

FinOps (Financial Operations) ek practice hai cloud spending ko optimize karne ki — Finance, Engineering aur Business teams milke kaam karein. Principles:

- Teams responsible for their cloud usage
- Everyone collaborates: Finance, DevOps, Business
- Reports should be timely and accurate
- Value of cloud vs cost balance karo

Azure Cost Management, tagging, budgets, Advisor — sab FinOps toolkit ka hissa hain. Cloud-native organizations ke liye critical practice.

Q196. What is Azure Cost Management export feature?

Azure Cost Management se cost data export karo:

- Daily/monthly usage data CSV mein
- Azure Blob Storage mein automatically save hota hai
- Power BI Desktop se connect karo analysis ke liye
- Custom dashboards banao
- Long-term trend analysis karo

Enterprise organizations apni BI tools mein Azure cost data integrate karte hain custom reporting ke liye. Automatically scheduled exports set kar sakte ho.

Q197. Azure Advisor kya recommendations deta hai?

Azure Advisor 5 categories mein recommendations deta hai:

1. Cost: Right-size VMs, delete unused resources, RI purchase suggestions
2. Security: Security posture improve karo (Defender for Cloud integration)
3. Reliability: Availability improve karo
4. Performance: Latency aur speed optimize karo
5. Operational Excellence: Best practices follow karo

Free service. Regular check karo — actionable insights milti hain. High impact items pehle fix karo.

Q198. What is Azure Cost Anomaly Detection?

Azure Cost Management mein anomaly detection feature hai jo unusual spending patterns detect karta hai:

- AI/ML se normal spending patterns learn karta hai
- Sudden spike ya unusual pattern detect hone par alert bheja jaata hai
- Email ya action groups se notify karo

Use case: Developer ne accidentally 100 powerful VMs create kar diye — anomaly alert aayega before huge bill aaye. Proactive cost management ke liye excellent feature.

Q199. Azure mein unused resources kaise identify karein?

Azure Advisor □ Cost recommendations mein milenge:

- Zero CPU utilization VMs (shutdown candidates)
- Unattached managed disks
- Idle virtual network gateways
- App Service plans with no apps
- Public IP addresses not associated with any resource

Regularly review karo Azure portal mein. Tagging se environment identify karo (dev = regularly cleanup). Automation scripts from Azure Automation bhi help kar sakte hain scheduled cleanups ke liye.

Q200. Summarize Azure pricing aur cost management for AZ-900.

Key takeaways:

- Factors: Resource type, region, tier, bandwidth, licensing, commitment
 - Tools: Pricing Calculator (estimate), TCO Calculator (on-prem vs Azure), Cost Management (track spending)
 - Savings: Reserved Instances (40-72%), Spot VMs (90%), Hybrid Benefit, Auto-scale
 - Governance: Budgets + alerts, Tags + cost allocation, Azure Advisor recommendations
 - SLA: Uptime guarantee, service credits for breach, composite SLA multiply karo
- Cloud cost optimization ongoing process hai – set aur forget nahi.

Q201. What is Azure Service Health?

Azure Service Health 3 components mein divided hai:

1. Azure Status: Global Azure outages public page
2. Service Health: Aapki subscriptions ki services ki health personalized view
3. Resource Health: Individual resource ki health (VM running/unavailable)

Alerts configure karo – incident, maintenance, health advisories pe email ya action group notification.

Incident ke time: Root cause analysis (RCA) milti hai Microsoft se 72 hours ke andar.

Q202. What is Azure Monitor?

Azure Monitor Azure resources ki monitoring aur observability platform hai:

- Metrics: Numerical performance data (CPU %, memory, requests/sec)
- Logs: Detailed diagnostic data (Log Analytics workspace)
- Alerts: Threshold breach hone par notify/act
- Dashboards: Visual monitoring views
- Insights: Application Insights, VM Insights, Container Insights
- Workbooks: Interactive reports

All Azure services automatically metrics Azure Monitor ko send karte hain.

Q203. What is Log Analytics workspace?

Log Analytics workspace Azure Monitor ka component hai — logs store karta hai query ke liye. Kusto Query Language (KQL) se powerful queries likho. Multiple sources se logs collect karo: Azure resources, VMs (via agent), on-premises servers, applications. Insights (Application Insights, VM Insights) bhi Log Analytics par based hain. Microsoft Sentinel bhi Log Analytics use karta hai backend mein.

Q204. What is Application Insights?

Application Insights Azure Monitor ka feature hai web applications monitor karne ke liye:

- Request rates, response times, failure rates
- Dependency tracking (DB calls, external APIs)
- Exception tracking
- User behavior (page views, sessions)
- Performance counters
- Custom events/metrics

SDK applications mein integrate karo ya agentless monitoring use karo. Developers ko production issues quickly identify karne mein help karta hai.

Q205. What are action groups in Azure Monitor?

Action Groups define karte hain kya hoga jab alert fire ho:

- Email/SMS/Push notifications
- Voice call
- Azure Automation runbook (script run karo)
- Azure Function (serverless action)
- Logic App (workflow trigger karo)
- ITSM tool (ServiceNow ticket create karo)
- Webhook (custom HTTP call)

Ek action group multiple alerts mein reuse kar sakte ho. Incident response automate karne ke liye powerful.

Q206. What is Azure Workbooks?

Azure Workbooks interactive reports banane ka tool hai — Azure Monitor, Log Analytics data se. Metrics, logs, parameters combine karo ek document mein. Parameterized — user input se filter karo. Templates available hain pre-built reports ke liye. Share karo team ke saath. Use cases: Security reports, performance reports, cost analysis dashboards. PowerBI ke bina complex analysis karo.

Q207. What are Azure Monitor alerts types?

Azure Monitor alert types:

1. Metric alerts: Value threshold breach kare (CPU > 80%)
2. Log search alerts: KQL query specific results return kare
3. Activity log alerts: Resource operations par (VM deleted, policy changed)
4. Service Health alerts: Azure outages/maintenance
5. Smart Detection: Application Insights mein anomaly auto-detect

Severity levels: 0 (Critical) to 4 (Verbose). Alert $\square$ Action Group trigger karo.

Q208. What is Azure Diagnostics settings?

Diagnostics Settings configure karte hain Azure resources kahan apne logs/metrics send karein:

- Log Analytics workspace (recommended)
- Azure Storage account (archive ke liye)
- Azure Event Hubs (third-party SIEM tools ke liye stream)
- Azure Monitor partner solutions

Har service ke specific log categories hoti hain enable karne ke liye. Enable karo needed categories sirf (cost control ke liye) – sab kuch unnecessarily enable mat karo.

Q209. What is KQL (Kusto Query Language)?

KQL (Kusto Query Language) Azure Log Analytics, Application Insights, aur Microsoft Sentinel mein use hoti hai. SQL-like but different syntax:

'''

AzureActivity

| where TimeGenerated > ago(24h)

| where ActivityStatus == 'Failed'

| summarize count() by Caller

| order by count_ desc

'''

Powerful pipe-based queries, easy to learn for data analysts. AZ-900 mein deep KQL nahi poochha jaata – sirf concept samajhna kaafi hai.

Q210. What is Azure Service Level Objective (SLO) vs SLA?

SLA (Service Level Agreement) – External commitment, contractual. Microsoft ka aapke saath agreement – breach hone par service credits.

SLO (Service Level Objective) – Internal target. Aapki team ka apna target – e.g., 'hum 99.95% uptime maintain karein.'

SLI (Service Level Indicator) – Actual measurement – e.g., uptime last 30 days 99.97% raha.

AZ-900 mein mostly SLA hi poochha jaata hai, SLO/SLI advanced concepts hain SRE ke liye.

Q211. What are preview services in Azure?

Preview services = features jo abhi testing phase mein hain:

- Public Preview: Koi bhi use kar sakta hai, production ke liye not recommended
- Private Preview: Invited customers only
- Generally Available (GA): Fully supported, SLA guaranteed

Previews SLA provide nahi karte. Pricing change ho sakti hai GA hone par. Azure Updates page par new features track karo. Exam ke liye: GA features SLA pe include hoti hain, previews nahi.

Q212. What is Azure Pay-Per-Use vs Flat Rate pricing?

Pay-Per-Use – Actual consumption ke hisaab se pay karo. Most flexible. Good for variable workloads. Azure Functions, Logic Apps, Storage API calls.

Flat Rate / Reserved – Fixed cost per month regardless of usage. Committed usage ke liye better ROI. VMs reserved, SQL database fixed tiers.

Azure ka default = pay-per-use (pay-as-you-go). Cost optimize karo reservations se jab usage predictable ho. Hybrid approach: Reserved base + pay-as-you-go peak ke liye.

Q213. What is Azure's commitment to sustainability?

Microsoft ka commitment:

- 2025 by: 100% renewable energy for data centers
- 2030 by: Carbon negative (remove more CO2 than emit)
- 2050 by: Remove all historical carbon emissions

Azure Emissions Insights tool se apna carbon footprint track karo. Sustainability Calculator available. Cloud shift karne se typically on-premises se 98% lower carbon emissions (shared infrastructure ki efficiency). ESG reporting ke liye Azure tools available hain.

Q214. What is the Azure free account validity?

Azure Free Account structure:

- Credit: \$200, valid 30 days. Expired ya credit exhausted hone par upgrade prompt.
- 12 months free services: Credit expire hone ke baad bhi 12 months tak specific services free.
- Always free: Indefinitely free specific services/limits.

After 30 days: Automatically Pay-as-you-go nahi hota — aapko manually upgrade karna hota hai. Student accounts: \$100 credit, 12 months, Azure for Students Starter bhi available (no credit card needed).

Q215. What is Azure Cost Allocation?

Cost Allocation Azure Cost Management mein ek feature hai jo shared costs distribute karne deta hai:

- Shared resources ka cost multiple departments mein split karo
- Tags ke basis par allocate karo
- Showback/Chargeback: Departments ko unka cost dikhao/bill karo

Example: Shared Azure Monitor workspace ka cost 3 teams mein equally split karo. Enterprise organizations mein IT departments cloud costs accurately allocate kar sakte hain business units ko.

Q216. What is the difference between Azure consumption and fixed pricing?

Consumption Pricing:

- Use karo tab pay karo
- No minimum
- Variable monthly bill
- Examples: Functions, Logic Apps, Storage transactions, Bandwidth

Fixed/Provisioned Pricing:

- Reserved capacity, fixed monthly cost
- Consistent bill regardless of actual use
- Better value if you always use it
- Examples: Reserved VMs, Azure SQL DTU-based, App Service plans

Most services offer both options — choose wisely based on usage patterns.

Q217. What is Azure Dev/Test pricing?

Azure Dev/Test pricing non-production environments ke liye discounted rates:

- Visual Studio subscribers ke liye special offers
- Windows VMs: Linux rates par (no Windows license cost)
- Specific services: Up to 50%+ discounts

Dev/Test subscription create karo Azure portal mein. Production ke liye eligible nahi — only for dev, testing, staging. Significant savings if large development teams have Azure resources running.

Q218. What happens if Azure SLA is breached?

SLA breach hone par:

- Service Credit milta hai – billing discount future invoice par
- Amount depends on downtime severity:
- < 99.9% uptime: 10% credit
- < 99% uptime: 25% credit
- < 95% uptime: 100% credit
- Credit automatically nahi milta – aapko claim karna hota hai (ticket raise karo)
- Time limit: Usually 2 months ke andar claim karo

SLA cash refund nahi, sirf Azure credits hain.

Q219. What is Azure Pricing Model for Kubernetes (AKS)?

AKS Control Plane:

- Standard tier: Small monthly fee (recently introduced)
- Free tier: No charge for control plane (limited SLA)

Worker Nodes:

- Pay karo VMs ke liye jo node pools mein hain
- Reserved Instances use karo node VMs ke liye savings ke liye
- Spot node pools (cheap, eviction tolerant workloads ke liye)

Additional costs: Storage (persistent volumes), Load Balancers, Container Registry (ACR), Bandwidth.

Q220. Summarize Azure Cost Management key concepts.

Summary:

- Pricing factors: Type, region, tier, bandwidth, licensing, commitment
 - Cost tools: Pricing Calculator, TCO Calculator, Cost Management + Billing, Azure Advisor
 - Savings strategies: Reserved Instances, Hybrid Benefit, Spot VMs, right-sizing, auto-scale, dev/test pricing
 - Governance: Budgets, tags, cost allocation, anomaly detection
 - Monitoring: Cost analysis, exports, Power BI integration
 - SLA: Uptime guarantee, service credits for breach, composite SLA multiply karein
- FinOps culture adopt karo continuous optimization ke liye.

Section 7: Azure Governance & Compliance

Q221. Azure Governance kya hota hai?

Azure Governance = Azure environments ko controlled, compliant aur well-managed rakhne ki practices aur tools. Bina governance ke: teams different naming conventions use karen, excess costs hon, security gaps hon, compliance violations hon. Governance components:

- Azure Policy: Rules enforce karo
- RBAC: Access control
- Blueprints: Standard environments
- Management Groups: Hierarchy organize karo
- Tagging: Resources label karo
- Cost Management: Spending control karo

Q222. Azure Policy effects kya hote hain?

Azure Policy 'effects' batate hain policy violation hone par kya hoga:

- Audit: Log karo, kuch block mat karo (non-compliant report)
- AuditIfNotExists: Agar related resource nahi hai toh audit karo
- Deny: Resource create/modify mat hone do
- Append: Properties add karo (tags, settings)
- Modify: Tags ya properties change karo
- DeployIfNotExists: Related resource deploy karo agar nahi hai
- Disabled: Policy temporarily disable karo

Q223. Azure Policy Initiative kya hota hai?

Initiative (Policy Set) multiple related policies ko ek group mein bundle karta hai. Example: 'ISO 27001' initiative mein 100+ individual policies hain. Ek baar assign karo sari policies ek saath lagu ho jaayein. Azure built-in initiatives: CIS Azure Benchmark, NIST SP 800-53, PCI DSS. Custom initiatives bhi banao. Simplifies compliance management at scale.

Q224. What is Azure Management Hierarchy?

Azure ka management hierarchy top to bottom:

1. Azure AD Tenant (root)
2. Root Management Group
3. Management Groups (custom, nested)
4. Subscriptions
5. Resource Groups
6. Resources

Policies, RBAC, Blueprints apply karo higher level par — automatically inherit karte hain neeche tak. Enterprise organizations is hierarchy ko carefully design karte hain governance ke liye.

Q225. What is Azure Landing Zone?

Azure Landing Zone ek pre-configured, scalable cloud environment hai jo enterprise Azure adoption ke liye ready hai. Include karta hai: networking hub, security, identity, management, governance sab already configured. Microsoft ke 'Enterprise Scale' architecture se guided hai. CAF (Cloud Adoption Framework) se aligned. New subscriptions ke liye starting point hai jo already all best practices follow karta ho.

Q226. Cloud Adoption Framework (CAF) kya hai?

Microsoft Cloud Adoption Framework Azure mein move karne ke liye proven guidance hai:
Phases:

1. Strategy: Business case define karo
2. Plan: Adoption plan banao
3. Ready: Landing zone prepare karo
4. Adopt: Migrate ya innovate
5. Govern: Governance establish karo
6. Manage: Operations ongoing
7. Secure: Security posture

Free documentation hai docs.microsoft.com par. Enterprises is framework follow karte hain large-scale cloud migrations ke liye.

Q227. Well-Architected Framework kya hai?

Azure Well-Architected Framework (WAF) 5 pillars par Azure solutions design karne ki guidance deta hai:

1. Reliability: Failures se recover karo
2. Security: Data aur systems protect karo
3. Cost Optimization: Value for money
4. Operational Excellence: Processes aur monitoring
5. Performance Efficiency: Workloads efficiently chalao

WAF Review assessment available hai — apna architecture evaluate karo aur recommendations pao. Solutions architects ke liye essential reference.

Q228. What is Microsoft Trust Center?

Microsoft Trust Center (microsoft.com/trust-center) Microsoft ka transparency portal hai:

- Privacy policies
- Compliance certifications (90+)
- Security practices
- Government data requests information
- Data residency details

Customers aur auditors yahan confirm karte hain ki Microsoft aur Azure certain standards comply karte hain. AZ-900 exam mein: Trust Center = compliance documentation ka source.

Q229. What are Azure compliance offerings?

Azure 90+ compliance certifications maintain karta hai:

Global: ISO 27001, ISO 27017, ISO 27018, SOC 1/2/3

US: FedRAMP, DoD IL2-IL6, HIPAA, HITRUST

Europe: GDPR, ENS (Spain), C5 (Germany)

Asia: IRAP (Australia), G-Cloud (UK), MTCS (Singapore), K-ISMS (Korea)

India: MeitY (MCA51), RBI, SEBI guidelines follow karta hai

Compliance Manager se check karo specific regulations ke liye aapki compliance status.

Q230. GDPR kya hai aur Azure iske saath kaise deal karta hai?

GDPR (General Data Protection Regulation) EU ka data privacy law hai — 2018 se effective.

Requirements:

- User consent data collection ke liye
- Right to be forgotten (data delete)
- Data breach notification 72 hours mein
- Data Protection Officer appoint karo

Azure GDPR ke liye:

- Data residency guarantee — EU mein rakhne ki option
- Compliance Manager mein GDPR assessment
- Microsoft Data Processing Agreement (DPA) provide karta hai
- Azure customers data controllers hain, Microsoft data processor.

Q231. What is Azure Resource Naming Conventions?

Naming conventions governance ka important hissa hain:

- Consistency: sab teams same pattern follow karein
- Example pattern: {resource-type}-{project}-{environment}-{region}-{instance}
- Example: vm-webapp-prod-eastus-001

Best practices:

- Lowercase use karo (case-sensitive resources ke liye)
- Hyphens for separators
- Abbreviations for resource types (vm, rg, st, kv)

Azure Policy se naming patterns enforce karo. Azure Naming Tool bhi available hai.

Q232. What is tagging strategy in Azure governance?

Effective tagging strategy:

Required tags (Azure Policy se enforce karo):

- Environment: dev/staging/prod
- Owner: team/person email
- CostCenter: billing code
- Application: app name
- CreatedBy: automation/person

Optional tags:

- ExpiryDate: temporary resources ke liye
- Compliance: HIPAA/PCI/etc.

Tag inheritance ARM templates mein set karo. Regular audits: tagging compliance check karo.

Q233. What is Azure Resource Graph?

Azure Resource Graph ek service hai jo Azure resources efficiently query karne deta hai — thousands of resources across subscriptions ek query mein. KQL use karta hai. Real-time data (not delayed). Use cases:

- Inventory management
- Compliance reporting
- Security investigations
- Cost analysis

Example: 'Find all VMs without specific tag in all subscriptions.' Azure portal mein Resource Graph Explorer available hai free mein.

Q234. What is the difference between Azure Policy and Initiative?

Azure Policy = single rule. Example: 'Allowed locations: East US, West US only.'

Initiative = Policy ka collection/bundle. Example: 'CIS Azure Benchmark' = 180+ individual policies.

Kab kya use karein:

- Simple single rule = Policy
- Related policies together apply karna = Initiative
- Compliance standards implement karna = Built-in Initiative (NIST, PCI, ISO, etc.)

Initiatives simplify large-scale governance — ek assign karo, 100 policies apply hongy.

Q235. What is Azure Cost Governance?

Cost Governance = spending control karna — rules aur processes se overspending rokna:

1. Budgets: Spending limits + alerts
2. Azure Policy: Expensive resources ban karo (e.g., GPU VMs allowed nahi)
3. Tags: Department-wise tracking
4. Management Group policies: Central control
5. Subscription segregation: Dev/Prod alag billing
6. Regular reviews: Advisor recommendations implement karo
7. RI utilization: Reservations waste nahi honni chahiye

Q236. What is Azure Lighthouse?

Azure Lighthouse managed service providers (MSPs) ko multiple customer tenants ek jagah se manage karne deta hai. Benefits:

- Service providers: Client Azure environments manage karo apne portal se
- Customers: Granular access give karo provider ko
- Visibility: Cross-tenant resource view
- Governance: Consistent policies across customers

Enterprises bhi Lighthouse use karte hain multiple subscriptions centrally manage karne ke liye.

Q237. What is Azure Subscription design best practices?

Subscription design considerations:

- Workload separation: Prod/Dev/Test alag subscriptions
- Regulatory isolation: Compliance boundary per subscription
- Cost accountability: Department per subscription
- Scaling limits: Azure per-subscription service limits

Common patterns:

- Single subscription (small orgs)
- Production/Non-production split
- Business unit subscriptions
- Application subscriptions
- Environment-based (Dev/Staging/Prod) subscriptions

Management Groups se govern karo multiple subscriptions.

Q238. What are Azure Regulatory Compliance dashboard?

Microsoft Defender for Cloud mein Regulatory Compliance dashboard:

- Multiple standards simultaneously monitor karo (PCI DSS, ISO 27001, CIS, NIST)
- Overall compliance score per standard
- Passing/failing controls breakdown
- Recommendations for failed controls
- Download compliance report (PDF/CSV) for auditors

Audit preparation ke liye very useful. Controls status real-time track karo. Policies enable karo specific assessments ke liye.

Q239. What is Azure Subscription limits and quotas?

Azure mein per-subscription service limits hain (quotas):

- VMs: 20,000 per subscription per region (default)
- Public IPs: 1,000 per subscription per region
- VNets: 1,000 per subscription per region

Limits approach hone par:

- Azure support se quota increase request karo
- Multiple subscriptions use karo across multiple regions

Enterprise organizations yahan plan karte hain. AZ-900 ke liye: quotas exist karte hain, increase possible hai via support.

Q240. What is Azure Sovereign Cloud compliance?

Azure Sovereign Clouds specific government requirements ke liye hain:

- Azure Government (US): FedRAMP High, DoD IL5/IL6, ITAR, IRS 1075
- Azure China: Chinese cybersecurity law compliance, data in China

Regular Azure commercial cloud se completely separate. Dedicated hardware, physically isolated. Government agencies aur their contractors use karte hain. Data residency guarantee sovereign cloud mein zyada strict hoti hai.

Q241. What is Shared Responsibility in Compliance?

Compliance mein bhi shared responsibility hota hai:

Microsoft ki responsibility:

- Physical data center security
- Infrastructure compliance certifications
- Platform-level controls

Customer ki responsibility:

- Data classification
- Access management
- Application security
- Regulatory controls (HIPAA: PHI protection aap karo)

Compliance Manager se customer responsibilities identify karo aur implement karo. 'Cloud mein hain toh compliant hain' yeh assumption galat hai.

Q242. What is Azure Policy compliance reporting?

Azure Policy Compliance portal mein:

- Overall compliance percentage
- Per-policy compliance
- Non-compliant resources list
- Remediation tasks (auto-fix non-compliant resources)
- Compliance history over time
- Export compliance data

Az-900 mein: Policies audit ya deny kar sakti hain. Remediation se existing resources fix ho sakte hain. DeployIfNotExists policy se missing configurations automatically deploy ho sakti hain.

Q243. What is Azure Cost governance through Policy?

Azure Policy se cost governance:

- 'Allowed VM SKUs': GPU VMs ban karo
- 'Allowed locations': Expensive regions ban karo
- 'Required tags': CostCenter tag mandatory
- 'Max allowed resource zones': Limits set karo
- 'Azure Defender must be enabled': Security spend mandate karo

Deny effect se expensive resources create hi na ho sakein. Development teams ko guardrails dene ke liye effective approach. Finance + IT collaboration needed.

Q244. What is Azure Blueprints vs ARM Templates?

ARM Templates – Infrastructure as code. Resources deploy karne ke liye JSON/Bicep files. Sirf deployment.

Azure Blueprints – Governance package. Bundle karo: ARM Templates + RBAC + Policies + Resource Groups. Track karta hai kaunsa subscription blueprint ka use kar raha hai. Version control. Compliance enforced.

Difference: ARM Template deploy karta hai, Blueprint govern bhi karta hai. Blueprint mein ARM Templates include ho sakte hain ek component ke roop mein.

Q245. What is Azure Policy exemption?

Policy Exemption specific resources ya scopes ko policy se exclude karne ka tarika hai:

- Waiver: Known non-compliance, accepted risk
- Mitigated: Alternative control in place

Exemptions expire dates set karo – temporary hona chahiye.

Audit trail maintain hota hai. Use carefully – har exemption security/compliance risk represent karta hai. Approval workflow recommended hai exemptions ke liye sensitive policies par.

Q246. What is Azure Resource Locks best practices?

Resource Locks best practices:

- Production resource groups par CanNotDelete lock lagao
- Core infrastructure (VNETs, Express Route) par ReadOnly consider karo
- Automation scripts mein lock remove operation lock re-apply pattern use karo
- Lock inheritance: Parent resource group lock all resources inherit
- Monitor lock changes: Activity Log mein track karo
- Document karo kaunse locks hain aur kyun

Q247. What is the Azure Naming and Tagging governance approach?

Governance approach:

1. Define naming convention policy – document karo
 2. Azure Policy se enforce karo naming patterns (audit initially, then deny)
 3. Required tags define karo
 4. Azure Policy se required tags enforce karo
 5. Azure Resource Graph se regularly audit karo compliance
 6. Tag inheritance ARM/Bicep templates mein set karo
 7. Training: Teams ko governance standards educate karo
- Governance day 1 se shuru karo – baad mein fix karna bahut mushkil hota hai.

Q248. What is Microsoft Privacy Policy for Azure?

Microsoft Azure privacy commitments:

- Customer data = aapka data. Microsoft advertise nahi karta.
- Government requests: Transparency report publish hoti hai
- Data residency: Choose karo data kahan store ho
- Data transfer: Adequate protection for transfers
- Right to access/delete customer data

Microsoft Online Services Terms (OST) aur Data Processing Agreement (DPA) formal documents hain jo aapke rights define karte hain. Trust Center par available hain.

Q249. What is Azure ISO certifications?

Azure ISO Certifications:

- ISO 27001: Information Security Management System
- ISO 27017: Cloud-specific security controls
- ISO 27018: PII in public cloud protection
- ISO 27701: Privacy Information Management System
- ISO 22301: Business Continuity Management
- ISO 9001: Quality Management

Third-party auditors (BSI, Bureau Veritas) annually audit Azure aur certifications renew karte hain. Customers unhe apne compliance documentation mein use kar sakte hain.

Q250. What is Azure SOC compliance?

SOC (System and Organization Controls) reports:

- SOC 1: Financial reporting controls
- SOC 2: Security, availability, processing integrity, confidentiality, privacy
- SOC 3: Public version of SOC 2 (no confidential details)

Type 1: Point-in-time assessment

Type 2: 6-12 month period assessment (more rigorous)

Azure SOC 2 Type 2 report available hai under NDA from Service Trust Portal. Customers use karte hain yeh reports apne auditors ko show karne ke liye.

Q251. What is Service Trust Portal?

Service Trust Portal (servicetrust.microsoft.com) Microsoft ka portal hai compliance documentation ke liye:

- SOC reports
- ISO certificates
- Penetration test results
- GDPR documents
- Industry-specific compliance docs

Register karo Microsoft account se. Documents download karo NDA accept karne ke baad. Security/compliance teams regularly check karte hain yahan. Customers apne vendors/auditors ko Azure compliance evidence share karne ke liye use karte hain.

Q252. What is Azure Compliance Manager assessment?

Compliance Manager mein assessments:

1. Ek regulation select karo (e.g., HIPAA)
2. Assessment milta hai all controls ke saath
3. Each control pe:
 - Microsoft-managed actions: Already done by Microsoft
 - Customer-managed actions: Aapko implement karna hai
4. Evidence upload karo
5. Compliance score track karo
6. Report export karo

Audit preparation drastically fast hoti hai Compliance Manager se. Gap analysis easily possible hai.

Q253. What is Azure Guest Configuration?

Azure Guest Configuration (ab Azure Machine Configuration) VM ke andar OS settings, installed software, aur configurations audit aur enforce karta hai:

- Windows: Registry settings, services, local users, installed software
- Linux: File permissions, installed packages, running services

Azure Policy ke extension ke roop mein kaam karta hai. Arc-enabled servers (on-premises bhi) support karta hai. Compliance reports mein include hota hai VM configuration compliance.

Q254. What is Azure Update Manager?

Azure Update Manager (pehle Update Management) VMs ke OS updates manage karta hai:

- Patch status view karo
- Schedule maintenance windows
- Immediately patch karo
- Compliance reports
- Azure VMs + Arc-enabled on-premises VMs

Security patching governance ke liye important. Unpatched systems = compliance violations. Policy se mandate karo updates aur Azure Update Manager se track karo implementation.

Q255. What is Azure Backup governance?

Azure Backup governance best practices:

- Azure Policy se enforce karo VMs backed up honi chahiye
- Backup Center se centralized monitoring
- Soft delete enable karo (14-day recovery window)
- RBAC: Backup operator alag role, delete nahi kar sakta
- Immutable vaults: Backup policy change/delete nahi ho sakti
- Cross-region restore: Secondary region se recovery test karo
- Regular DR drills karo backup restore test karne ke liye

Q256. What is Azure Defender for DevOps?

Microsoft Defender for DevOps (Defender for Cloud component) DevOps pipelines secure karta hai:

- GitHub aur Azure DevOps ke saath integrate
- Code scanning: Vulnerabilities, secrets in code detect karo
- IaC scanning: ARM/Bicep/Terraform misconfigurations
- Container image scanning: Base images vulnerabilities
- Pull request annotations: Developer ko feedback code review mein

Shift-left security: Security issues production se pehle developer stage par pakdo.

Q257. What is Microsoft Defender for Cloud regulatory compliance?

Defender for Cloud mein Regulatory Compliance:

- Multiple standards simultaneously monitor karo
- Automated control assessments
- Manual control attestation
- Continuous monitoring — new resources automatically assessed
- Audit reports download karo
- Workflow automation: Non-compliant resources par automatically tickets create karo

Enterprise compliance teams daily dashboard check karte hain aur remediation assign karte hain teams ko.

Q258. What is Azure Governance maturity model?

Governance Maturity evolves over time:

Level 1 – Basic: Manual processes, no policies, individual subscriptions

Level 2 – Defined: Naming conventions documented, some policies, tagging started

Level 3 – Managed: Policies enforced, RBAC properly configured, cost tracking

Level 4 – Optimized: Blueprints/Landing zones, automated governance, compliance dashboards, FinOps

Cloud Adoption Framework guide karta hai organizations is maturity journey par. Start karo basic se, evolve karte jao.

Q259. What is Azure Confidential Computing?

Azure Confidential Computing data ko use mein bhi protect karta hai – typically data encrypted at rest aur in transit hoti hai, lekin memory mein processing ke time unencrypted hoti thi. Confidential Computing TEE (Trusted Execution Environment) use karta hai – data encrypted rehti hai processing ke time bhi. Use cases: Multi-party computation, healthcare data analysis, financial data. Azure confidential VMs aur Confidential containers available hain.

Q260. Summarize Azure Governance and Compliance for AZ-900.

Key topics:

- Azure Policy: Rules enforce karo (Deny, Audit, Deploy)
- RBAC: Access control (Principal + Role + Scope)
- Management Groups: Subscription hierarchy
- Blueprints: Governance packages
- Resource Locks: Accidental changes rokna
- Tagging: Cost tracking, metadata
- Compliance tools: Compliance Manager, Trust Center, Service Trust Portal
- Compliance standards: ISO, SOC, GDPR, HIPAA, FedRAMP, PCI DSS
- Cost governance: Policies for resources + budgets + tagging

Governance = proactive; compliance = continuous verification.

Section 8: Azure Tools, Monitoring & Additional Services

Q261. Azure Monitoring tools ka overview deta hoon.

Azure monitoring ecosystem:

- Azure Monitor: Core platform – metrics, logs, alerts
- Log Analytics: Log storage + KQL queries
- Application Insights: Application performance monitoring
- Azure Service Health: Azure platform health
- Azure Advisor: Best practice recommendations
- Microsoft Defender for Cloud: Security monitoring
- Azure Network Watcher: Network diagnostics
- Azure Sentinel: Security event management

Ye sab together complete observability dete hain – infrastructure se application tak.

Q262. What is Azure Automation?

Azure Automation repetitive tasks automate karne ke liye service hai:

- Runbooks: PowerShell ya Python scripts cloud mein run karo
- DSC (Desired State Configuration): VM configurations enforce karo
- Update Management: OS patches automate karo
- Change Tracking: VMs mein changes track karo
- Process Automation: Scheduled ya event-triggered scripts

Use cases: VM start/stop schedule karo, resource cleanup karo, reporting automate karo. Significant OpEx savings possible hai automation se.

Q263. What is Azure DevTest Labs?

Azure DevTest Labs development aur testing environments quickly create karne ke liye service hai:

- Self-service environments: Developers khud VMs create kar sakein
- Cost control: Auto-shutdown, max VMs per user, allowed VM sizes
- Reusable templates: Artifacts, formulas, custom images
- Claimable VMs: Pre-created VMs pool maintain karo

Development teams ke liye cloud environments wastage rokne mein help karta hai. No separate cost – sirf underlying resources pay karo.

Q264. What is Azure Migrate?

Azure Migrate on-premises workloads Azure mein migrate karne ka hub hai:

- Discovery & Assessment: On-premises VMs discover karo, Azure readiness assess karo, cost estimate karo
 - Server Migration: VMware, Hyper-V, physical servers migrate karo
 - Database Migration: SQL Server, MySQL, PostgreSQL migrate karo
 - Web App Migration: ASP.NET apps App Service mein migrate karo
 - Virtual Desktop Migration: VDI solutions migrate karo
- Free tool, centralized hub. Partner tools bhi integrate hote hain.

Q265. What are Azure migration strategies (6Rs)?

Migration strategies:

1. Rehost (Lift & Shift): As-is Azure mein move karo, minimal changes
 2. Refactor: Minor code changes, Platform better utilize karo (App Service ke liye)
 3. Rearchitect: Significantly modify karo cloud-native features use karne ke liye
 4. Rebuild: Cloud-native application scratch se banao
 5. Replace: SaaS se replace karo (custom app $\square$ Microsoft 365)
 6. Retire: Decommission karo — nahi chahiye
- Most organizations mix use karte hain. Lift & Shift fastest, Rebuild most cloud-optimized.

Q266. What is Azure App Configuration?

Azure App Configuration application settings aur feature flags centrally manage karne ke liye service hai:

- Config values: Connection strings, API URLs, timeouts centrally store karo
- Feature flags: Code mein features toggle on/off karo bina redeployment ke
- Dynamic refresh: Running applications ko naye config values mil jaayein
- Point-in-time snapshot: Config history
- Geo-replication: Multiple regions mein sync

Microservices environments mein multiple services ka configuration ek jagah manage karna easy ho jaata hai.

Q267. What is Azure Event Grid?

Azure Event Grid event routing service hai — events publishers se subscribers tak deliver karo reliably:

- Publishers: Azure services (Blob storage upload, VM created, Policy violated, etc.) ya custom apps
 - Subscribers: Azure Functions, Logic Apps, Event Hubs, Webhooks
- Event-driven architecture ke liye. Near-real-time, low latency. Pay per event. Example: Image upload hone par Function trigger karo jo thumbnail banaye. Millions of events/second handle kar sakta hai.

Q268. What is Azure Event Hubs?

Azure Event Hubs big data streaming platform hai — millions of events per second ingest karo:

- Real-time data ingestion
- IoT telemetry, clickstream data, logs
- Consumers: Stream Analytics, Databricks, custom apps
- Kafka compatible (Kafka apps bina changes ke connect kar sakte hain)
- Capture: Raw events Azure Storage/ADLS mein store karo

Event Grid se alag: Event Hubs high-throughput data streaming ke liye, Event Grid event routing ke liye (routing focus, not high-volume streaming).

Q269. What is Azure Service Bus?

Azure Service Bus enterprise messaging service hai:

- Queues: One-to-one messaging (producer ⇄ consumer)
- Topics + Subscriptions: One-to-many messaging (publisher ⇄ multiple subscribers)
- Guaranteed delivery, ordering, dead-letter queue
- Session handling, scheduled messages, deferred processing

Enterprise integration ke liye. Zyada advanced than Azure Queue Storage (which is simpler, cheaper). Use karo jab complex messaging patterns chahiye — ordering, transactions, sessions, etc.

Q270. What is Azure Notification Hubs?

Azure Notification Hubs push notifications mobile applications ke liye — millions of devices tak. Supported: Apple (APNs), Google (FCM), Windows (WNS), Amazon (ADM), Baidu.

Use case: Banking app: transaction alert push karo sab users ko. News app: breaking news push karo. Simple API: Ek call karo, Notification Hubs sab platforms tak deliver kare. Millions of devices tak deliver karo minutes mein.

Q271. What is Azure IoT Hub?

Azure IoT Hub IoT (Internet of Things) devices manage karne aur communicate karne ke liye managed service hai:

- Device-to-cloud: Devices se data cloud mein aata hai
- Cloud-to-device: Cloud se devices ko commands jaate hain
- Device registry: Sab devices track karo
- Authentication: Per-device keys ya certificates
- Message routing: Event Grid ya Event Hubs par forward karo

Factory sensors, smart home, connected cars, healthcare devices — sab IoT Hub se connect ho sakte hain.

Q272. What is Azure Cognitive Services / Azure AI Services?

Azure AI Services (pehle Cognitive Services) pre-built AI APIs hain — ML expertise nahi chahiye:

- Vision: Image/video analysis, face detection, OCR
- Speech: Speech-to-text, text-to-speech, translation
- Language: Text analysis, sentiment, key phrases, translation
- Decision: Anomaly detection, content moderation, personalizer
- OpenAI Service: GPT-4, DALL-E, Whisper

REST APIs ya SDKs se call karo. Pay-per-call pricing. Use karke AI capabilities quickly add karo apps mein.

Q273. What is Azure Machine Learning?

Azure Machine Learning (AML) ML models train, deploy aur manage karne ka complete platform hai:

- Automated ML (AutoML): Best algorithm automatically find karo
- Designer: Drag-and-drop ML pipelines
- Notebooks: Jupyter notebooks cloud mein
- MLflow integration: Experiment tracking
- Model registry: Models version control
- Deployment: Real-time (AKS) ya batch inference
- Responsible AI: Fairness, explainability, error analysis

Data scientists ke liye enterprise-grade ML platform.

Q274. What is Azure Synapse Analytics?

Azure Synapse Analytics ek unified analytics platform hai:

- Data Warehouse: Azure SQL Data Warehouse (serverless ya dedicated pool)
- Big Data: Apache Spark integration
- Data Integration: Azure Data Factory pipelines
- Power BI integration: Direct reporting
- Data Explorer: Time series analytics

Ek platform mein: Ingest □ Store □ Prepare □ Model □ Serve. Data engineers, data scientists, business analysts sab use kar sakte hain. Organizations jo large-scale analytics karte hain unke liye.

Q275. What is Azure Data Factory?

Azure Data Factory (ADF) cloud-native ETL (Extract, Transform, Load) aur data integration service hai:

- 90+ connectors: SQL, Salesforce, S3, REST APIs, SFTP, SAP, etc.
- Data pipelines visually design karo
- Data transformation (mapping data flows)
- Schedule aur trigger pipelines
- Monitoring aur alerting

Data engineers use karte hain raw data ko data warehouse mein move karne ke liye. Azure Synapse Analytics mein integrated bhi hai.

Q276. What is Azure Databricks?

Azure Databricks Apache Spark based analytics platform hai — Microsoft aur Databricks ka collaboration:

- Big data processing
- Machine learning
- Real-time streaming
- SQL analytics
- Collaborative notebooks

Azure-native integrations: Azure Storage, Synapse, ADF, Key Vault, Active Directory. Data engineering aur data science teams use karte hain. Petabyte scale data process karo cluster par. Premium features: Delta Lake, MLflow, Unity Catalog.

Q277. What is Azure Stream Analytics?

Azure Stream Analytics real-time data stream processing service hai:

- Input: Event Hubs, IoT Hub, Blob Storage
- SQL-like queries: Filter, aggregate, join streams
- Output: Power BI, storage, databases, Functions

Use cases: Real-time dashboards, anomaly detection, live leaderboards, fraud detection.

Example: 10,000 IoT sensors data baar baar bhejte hain □ Stream Analytics avg temperature calculate karo per minute □ Power BI dashboard update karo real-time.

Q278. What is Azure Power BI?

Power BI business intelligence tool hai — data se insights visualize karo:

- Reports aur dashboards
- Multiple data sources connect karo
- Azure services se native integration
- Mobile app
- Sharing aur collaboration

Power BI Embedded: Azure service hai jo Power BI reports apne applications mein embed karne ke liye. Power BI + Azure Synapse + ADF: Complete analytics solution banana ke liye yeh combination common hai.

Q279. What is Azure Blockchain Service? (Deprecated)

Azure Blockchain Service ek managed blockchain ledger service tha – Quorum/Ethereum based. 2021 mein deprecated hua. Alternatives:

- ConsenSys Quorum on Azure VMs
- Azure SQL Ledger (tamper-evident tables – blockchain se inspired, centralized)
- Hyperledger Fabric on AKS

Az-900 perspective: Azure services evolve hote hain – kuch deprecated ho jaate hain. Current Azure catalog check karo. Azure SQL Ledger zyada commonly used tamper-evident solution hai aaj.

Q280. What is Azure Quantum?

Azure Quantum ek quantum computing cloud service hai:

- Multiple quantum hardware providers: IonQ, Quantinuum, Rigetti
- Quantum simulators bhi available
- Q# programming language (Microsoft ka quantum language)
- Optimization solvers (classical + quantum)

Aaj quantum computers classical computers se faster nahi hain most tasks ke liye – specialized problems ke liye (cryptography, optimization, chemistry simulation) potential hai. Early-stage technology, active research area.

Q281. What is Azure Applied AI Services?

Azure Applied AI Services specific business scenarios ke liye pre-built AI solutions hain:

- Azure Form Recognizer (Document Intelligence): Documents se data extract karo
- Azure Bot Service: Intelligent chatbots banao
- Azure Cognitive Search: AI-powered search
- Azure Video Indexer: Video content analysis
- Azure Metrics Advisor: Anomaly detection for time-series data
- Azure Immersive Reader: Reading comprehension improvement

Generic AI APIs (Cognitive Services) se zyada specific business use cases ke liye ready solutions.

Q282. What is Azure OpenAI Service?

Azure OpenAI Service GPT-4, GPT-3.5, DALL-E, Whisper, Embeddings models ko enterprise customers ke liye provide karta hai:

- Enterprise security aur compliance
- Private deployment – aapka data model train karne ke liye use nahi hota
- Azure Active Directory integration
- VNet integration, private endpoints
- Content filtering
- SLA guarantees

Developers Azure OpenAI use karte hain apne applications mein generative AI add karne ke liye – chat, code generation, summarization, image generation, etc.

Q283. What is Azure Mixed Reality services?

Azure Mixed Reality:

- Azure Spatial Anchors: Shared AR experiences create karo — multiple devices same location dekh sakein
- Azure Remote Rendering: Complex 3D models cloud mein render karo, HoloLens/Mobile par stream karo
- Azure Object Anchors: Real-world objects detect karo aur align 3D models

Industrial use cases: Manufacturing assembly guidance, equipment maintenance, architecture visualization. Early-stage but growing niche market.

Q284. What is Azure Synapse Link?

Azure Synapse Link operational databases (Cosmos DB, Azure SQL, SQL Server, Dataverse) aur Azure Synapse Analytics ke beech real-time integration provide karta hai — bina ETL pipelines ke.

Kaise kaam karta hai: Changes automatically analytical store mein replicate hote hain. Production database impact nahi hota.

Use case: E-commerce transactions real-time analytics. OLTP workload chal raha hai, simultaneously OLAP analytics bhi — data copy karne ki zaroorat nahi.

Q285. What is Azure Health Data Services?

Azure Health Data Services healthcare data ke liye specialized platform hai:

- FHIR Service: Fast Healthcare Interoperability Resources API (medical records exchange standard)
- DICOM Service: Medical imaging data (CT scans, MRIs)
- MedTech Service: IoT medical devices se data ingest
- HIPAA compliant
- Azure AD integration

Hospitals, health tech companies patient data securely exchange kar sakti hain. EHR systems se integrate hota hai (Epic, Cerner).

Q286. What is Azure Communications Services?

Azure Communication Services developers ko communication features apps mein add karne deta hai:

- Voice/Video calling
- Chat
- SMS
- Email
- Teams interoperability
- Phone number provisioning

APIs provide karta hai — koi telephony infrastructure manage nahi karna. Microsoft Teams ka backbone same infrastructure use karta hai. Telehealth apps, customer service platforms, remote collaboration tools banao.

Q287. What is Azure Fluid Relay?

Azure Fluid Relay real-time collaborative experiences ke liye backend hai — jaise Google Docs ki real-time co-editing.

Fluid Framework (Microsoft ka open-source) ke saath kaam karta hai. Multiple users simultaneously data edit kar sakein, changes instantly sync hon. Use cases: Collaborative documents, real-time dashboards, multiplayer experiences, live presentations.

Q288. What is Azure SignalR Service?

Azure SignalR Service real-time web functionality add karne ke liye managed service hai:

- Server se clients ko push messages karo (nahi chahiye client polling)
- WebSockets, Long Polling, Server-Sent Events support
- Automatic scaling

Use cases: Live chat, real-time notifications, live sports scores, stock tickers, collaborative whiteboards. Traditional REST APIs mein client baar baar poll karta hai — SignalR se server push karta hai — efficient.

Q289. What is Azure Cosmos DB?

Azure Cosmos DB globally distributed, multi-model NoSQL database hai:

- Multiple APIs: SQL (Core), MongoDB, Cassandra, Gremlin (graph), Table
- Global distribution: 1-click se duniya bhar mein replicate karo
- Guaranteed SLAs: 99.999% availability
- Multiple consistency levels: 5 levels choose karo
- Serverless ya provisioned throughput

Use cases: IoT, gaming, retail catalogs, personalization, social media. Microsoft ke sabse flagship Azure database services mein se ek.

Q290. What is Azure SQL Database?

Azure SQL Database fully managed relational database hai (SQL Server based):

- PaaS: OS, patches, backups sab Azure manage karta hai
- Built-in HA: 99.99% SLA
- Automatic tuning, intelligent query processing
- Elastic pools: Multiple databases resources share karein
- Serverless: Auto-pause/resume, pay-per-use
- Hyperscale: Up to 100TB

On-premises SQL Server apps Azure mein migrate karne ke liye natural choice. Azure Database Migration Service se migrate karo.

Q291. What are Azure database services overview?

Azure relational databases:

- Azure SQL Database (SQL Server managed)
- Azure Database for PostgreSQL
- Azure Database for MySQL
- Azure Database for MariaDB

Azure NoSQL databases:

- Cosmos DB (multi-model)
- Azure Table Storage (key-value)
- Azure Cache for Redis (in-memory)

Azure analytics databases:

- Azure Synapse (data warehouse)
- Azure Databricks (big data)

Right database choose karo workload ke hisaab se — relational vs NoSQL vs analytics.

Q292. What is Azure Cache for Redis?

Azure Cache for Redis in-memory data store hai — database queries aur application performance accelerate karo:

- Session store: User sessions fast store karo
- Caching: Frequent database queries cache karo
- Message broker: Pub/Sub patterns
- Leaderboards: Sorted sets for gaming
- Distributed lock: Distributed applications ke liye

Sub-millisecond latency. Fully managed Redis. Tiers: Basic, Standard (replication), Premium (clustering, persistence, VNet). E-commerce, gaming, social media par commonly used.

Q293. What is Azure Logic Apps vs Azure Functions?

Azure Functions – Code-centric, developer writes code, any complex logic, stateless (or durable), per-execution pricing.

Azure Logic Apps – Low-code/no-code, visual designer, 400+ built-in connectors, business workflow focus, stateful by default.

Kab kya: Complex custom code □ Functions. Business process automation without coding □ Logic Apps. Enterprise integration (SAP, Salesforce, Oracle) □ Logic Apps. Both can call each other. Often used together in integration scenarios.

Q294. What is Azure API Management?

Azure API Management (APIM) APIs publish, secure, transform aur monitor karne ka gateway hai:

- API Gateway: All APIs ek endpoint se
- Rate limiting, quotas
- Authentication (API keys, OAuth, JWT)
- Request/response transformation
- Caching
- Developer portal: API documentation, testing
- Analytics: Who is calling which API

Organizations jo external developers ko APIs expose karte hain ya microservices internally manage karte hain unke liye essential tool.

Q295. What is Azure Static Web Apps?

Azure Static Web Apps static frontend applications (HTML, CSS, JavaScript, React, Vue, Angular) host karne ke liye free tier service hai:

- GitHub/Azure DevOps CI/CD built-in
- Global CDN distribution
- Custom domains + free SSL
- Authentication built-in (AAD, GitHub, Twitter, etc.)
- Azure Functions backend optional
- Free tier available for personal projects

Jamstack architecture ke liye ideal. Developers quickly production-ready apps deploy kar sakte hain zero configuration ke saath.

Q296. What is Azure Spring Apps?

Azure Spring Apps (pehle Azure Spring Cloud) managed service hai specifically Spring Boot aur Steeltoe (Java/.NET) microservices ke liye:

- Service discovery, config management, circuit breaker sab managed
- Developer sirf business logic likhein
- Zero-trust security
- Blue-green deployments
- Autoscaling
- Application monitoring built-in

Microsoft aur VMware collaboration. Java enterprises ke liye fastest Azure deployment path hai.

Q297. What is Azure Service Catalog (Managed Applications)?

Azure Managed Applications organizations ko standard solutions publish karne deti hain apni catalog mein:

- Internal: IT team approved apps publish karo, employees self-service deploy karein
- Azure Marketplace: ISVs managed apps sell karein

ManagedApplication = pre-configured Azure resources + managed by publisher. Customer ko management complexity nahi hoti. Enterprise use case: Approved DR solution, approved monitoring solution catalog mein rakho, teams deploy karein standard way mein.

Q298. What is Azure Resource Graph Explorer?

Azure Resource Graph Explorer portal tool hai powerful queries run karne ke liye across all resources:

- KQL queries use karo
- All subscriptions at once query karo
- Real-time results
- Save queries, share queries
- Sample queries available

Examples:

- 'All VMs not running tag X'
- 'All public IPs'
- 'Resources in region X'
- 'Compliance status of all resources'

Governance aur security teams ka daily tool. Portal mein free available.

Q299. What new services has Azure added recently (up to 2025)?

Recent Azure additions (2023-2025 approximate):

- Azure OpenAI Service GA: GPT-4, DALL-E enterprise access
- Microsoft Copilot for Azure: AI-powered management assistant
- Azure Kubernetes Fleet Manager: Multi-cluster Kubernetes management
- Azure Container Storage: Purpose-built container storage
- Azure Operator 5G Core: Telecom 5G infrastructure
- Azure Confidential VMs: AMD SEV-SNP based
- Microsoft Entra rebranding: Azure AD family renamed

Azure evolves rapidly – 100+ new features monthly. Azure Updates page track karo.

Q300. AZ-900 exam ke liye final summary aur tips.

AZ-900 Exam Tips — Akhilesh Chaudhari:

Exam Details: 40-60 questions, 60 min, 700/1000 passing score.

High Priority Topics:

1. Cloud concepts: IaaS/PaaS/SaaS, public/private/hybrid, CapEx/OpEx
2. Azure core services: Compute (VM, App Service, Functions), Storage (Blob, Files, Queue), Networking (VNet, NSG, VPN), Database
3. Security: Azure AD/Entra ID, RBAC, Key Vault, Defender for Cloud, Zero Trust
4. Governance: Policy, RBAC, Blueprints, Locks, Tags
5. Cost: Pricing Calculator, TCO, Reserved Instances, Advisor
6. SLA: Composite SLA, uptime percentages

Study Resources: Microsoft Learn (free), AZ-900 Practice Tests, Hands-on Azure Free Account.

Best of luck! — Akhilesh Chaudhari (@akhilesh.webdev)